Incidence-Guided Treatment Assignment Designs for Spatial Cluster Randomized Trials

Comprehensive Simulation Study Report

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2026-04-08

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1 Introduction

This report provides a comprehensive reference for the simulation study evaluating **8 treatment assignment designs** for Spatial Cluster Randomized Trials (CRTs) when outcome incidence is heterogeneous and spatially structured.

> *Given known historical incidence data exhibiting spatial heterogeneity, which treatment assignment strategy minimizes estimation error for the direct intervention effect $\hat{\tau}$ across a range of spatial dependence and spillover configurations?*

Application context: Sudden Unexpected Death (SUD) in North Carolina counties. The Poisson incidence mode uses an SUD base rate of 35 per 100,000, consistent with county-level analyses.

The study uses Maximum Likelihood Estimation via a spatial autoregressive model (MLE) as the primary estimator, evaluated across 2,560 parameter scenarios. Three analytical questions structure the results:

- **Q1:** Which design performs best within each incidence generation mode?
- **Q2:** How do design rankings change as individual parameters vary?
- **Q3:** Which design wins for specific parameter combinations?

---

2 Study Region and Spatial Structure

The study region is a regular square lattice grid:

$$\mathcal{G} = \{s_1, s_2, \dots, s_N\}$$

where each cluster s_i is located at integer coordinates (x_i, y_i) on a $G \times G$ grid, so $N = G^2$. In this study, $G = 10$ and $N = 100$.

2.1 Spatial Contiguity

Two contiguity definitions are considered:

- **Rook contiguity**: clusters sharing an edge (4-connected).

$$\mathcal{N}_R(s_i) = \{s_j : |x_i - x_j| + |y_i - y_j| = 1\}$$

- **Queen contiguity**: clusters sharing an edge or vertex (8-connected).

$$\mathcal{N}_Q(s_i) = \{s_j : \max(|x_i - x_j|, |y_i - y_j|) = 1\}$$

2.2 Spatial Weight Matrix

Let $\mathbf{A}$ denote the binary adjacency matrix where $A_{ij} = \mathbf{1}[s_j \in \mathcal{N}(s_i)]$. The row-standardized spatial weight matrix is:

$$W_{ij} = \frac{A_{ij}}{\sum_{k=1}^{N} A_{ik}}, \quad i \neq j; \quad W_{ii} = 0$$

This ensures $\sum_j W_{ij} = 1$ for each row, so spatial lags represent neighborhood averages. We construct $\mathbf{W}_R$ (rook) and $\mathbf{W}_Q$ (queen) weight matrices.

---

3 Outcome Data Generating Process

3.1 Spatial Durbin Model (SDM)

The outcome vector $\mathbf{Y} = (Y_1, \dots, Y_N)^\top$ is generated from the Spatial Durbin Model:

$$\mathbf{Y} = (\mathbf{I}_N - \rho\mathbf{W})^{-1} \left[\tau\mathbf{Z} + \gamma \cdot \text{Spill}(\mathbf{Z}) + \beta\mathbf{X} + \varepsilon\right]$$

where:

Symbol	Description	Role
$\mathbf{Y}$	$N \times 1$ outcome vector	Response
ρ	Spatial autocorrelation parameter	Controls global spatial dependence in $\mathbf{Y}$
$\mathbf{W}$	$N \times N$ row-standardized weight matrix	Spatial structure (rook or queen)
τ	Direct treatment effect	**Estimand of interest** (swept: $\{0.8, 1.0, 1.5, 2.0, 3.0\}$; primary: 1.0)
$\mathbf{Z}$	$N \times 1$ binary treatment vector	$Z_i \in \{0, 1\}$
γ	Spillover effect magnitude	Indirect effect via treated neighbors
$\text{Spill}(\mathbf{Z})$	$N \times 1$ spillover term	See below
β	Effect of baseline incidence on outcome	Covariate coefficient
$\mathbf{X}$	$N \times 1$ baseline incidence vector	$X_i \in [0, 1]$
ε	$N \times 1$ error vector	$\varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$

The spatial multiplier $(\mathbf{I}_N - \rho\mathbf{W})^{-1}$ induces global spatial dependence.

Stationarity condition: $|\rho| < 1/\lambda_{\max}(\mathbf{W})$. For row-standardized $\mathbf{W}$, $\lambda_{\max} = 1$, so $|\rho| < 1$.

3.2 Spillover Construction

$$\text{Spill}_i = \frac{\sum_{j \in \mathcal{N}(i)} Z_j}{|\mathcal{N}(i)|} = (\mathbf{WZ})_i$$

Two spillover types are considered:

1. **Control-only spillover**: $\text{Spill}_i^{(\text{ctrl})} = \gamma \cdot (\mathbf{WZ})_i \cdot (1 - Z_i)$
2. **Bidirectional spillover**: $\text{Spill}_i^{(\text{both})} = \gamma \cdot (\mathbf{WZ})_i$

4 Incidence Generation Modes

The baseline incidence vector $\mathbf{X}$ represents prior observed outcome rates. Three generation modes are considered, all producing values in $[0, 1]$.

4.1 Mode 1: Independent and Identically Distributed (iid)

$$X_i \overset{\text{iid}}{\sim} \text{Uniform}(0, 1), \quad i = 1, \dots, N$$

No spatial structure; serves as the baseline. One config ($\rho_X = \text{N/A}$).

4.2 Mode 2: Spatially Correlated (SAR Filter)

$$\mathbf{X}^* = (\mathbf{I}_N - \rho_X \mathbf{W})^{-1} \mathbf{u}, \quad \mathbf{u} \sim N(\mathbf{0}, \mathbf{I}_N)$$

$$X_i = \Phi(X_i^*)$$

where $\Phi(\cdot)$ is the standard normal CDF. Two configs: $\rho_X \in \{0.20, 0.50\}$.

4.3 Mode 3: Poisson Count-Based

1. Spatially correlated log-rates: $\log \lambda^* = \mu_0 \cdot \mathbf{1} + (\mathbf{I}_N - \rho_X \mathbf{W})^{-1} \mathbf{u}$
2. Poisson counts: $C_i \sim \text{Poisson}(\lambda_i^* \cdot P_i)$ where $P_i = 1{,}000$
3. Observed rates: $R_i = C_i / P_i$
4. Rank-normalized: $X_i = (\text{rank}(R_i) - 0.5)/N$

Base rate $\mu_0 = \log(35/100{,}000)$ for SUD. Two configs: $\rho_X \in \{0.20, 0.50\}$.

Critical design invariant: Incidence is generated **once per (mode, ρ_X) configuration** and held fixed across all (ρ, γ, nb_type) scenarios.

4.4 Incidence Spatial Patterns

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5 Treatment Assignment Designs

Eight designs are evaluated. Each produces a binary treatment vector $\mathbf{Z} \in \{0, 1\}^N$.

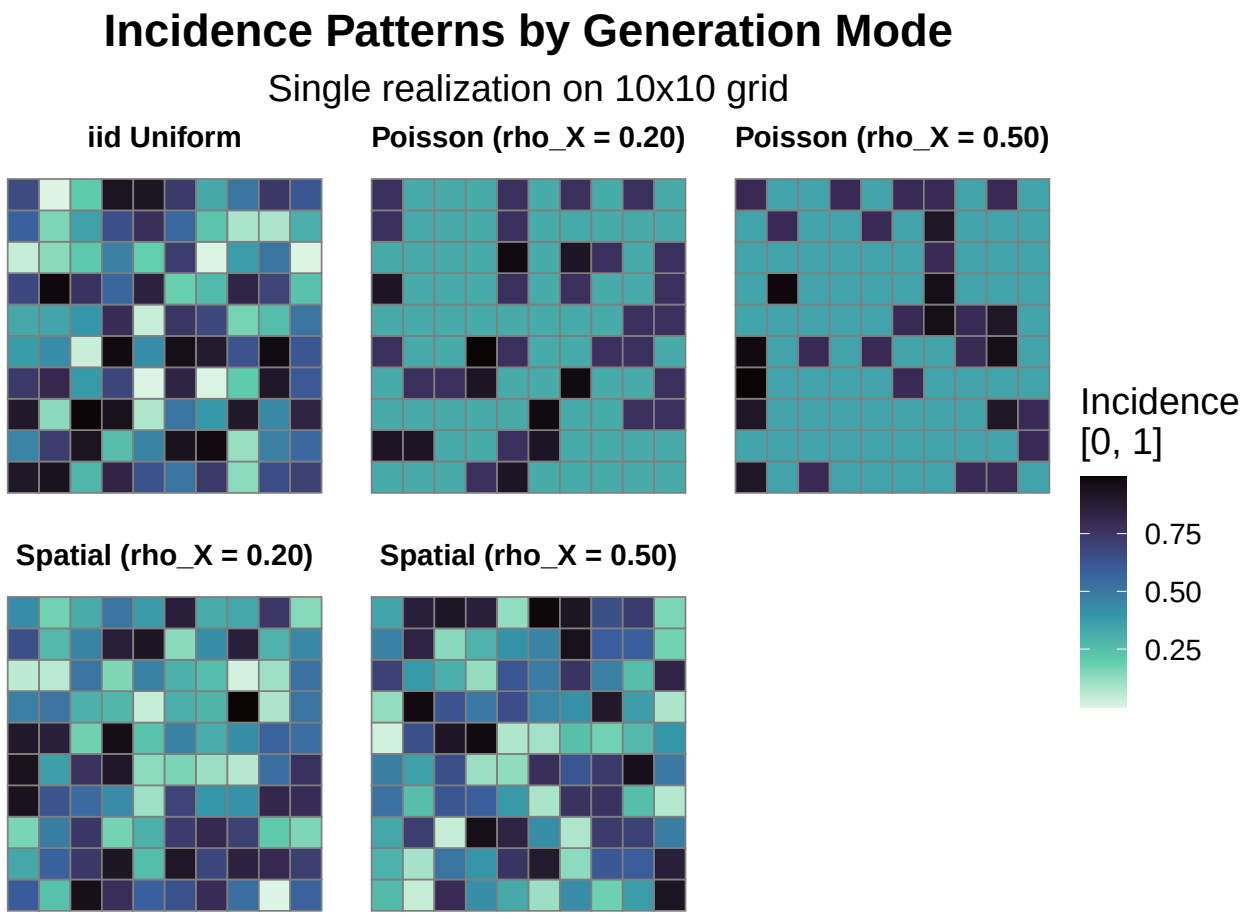

Figure 1: Incidence spatial patterns across generation modes. Color represents incidence (darker = higher burden). Spatial and Poisson modes show visible spatial clustering absent from iid.

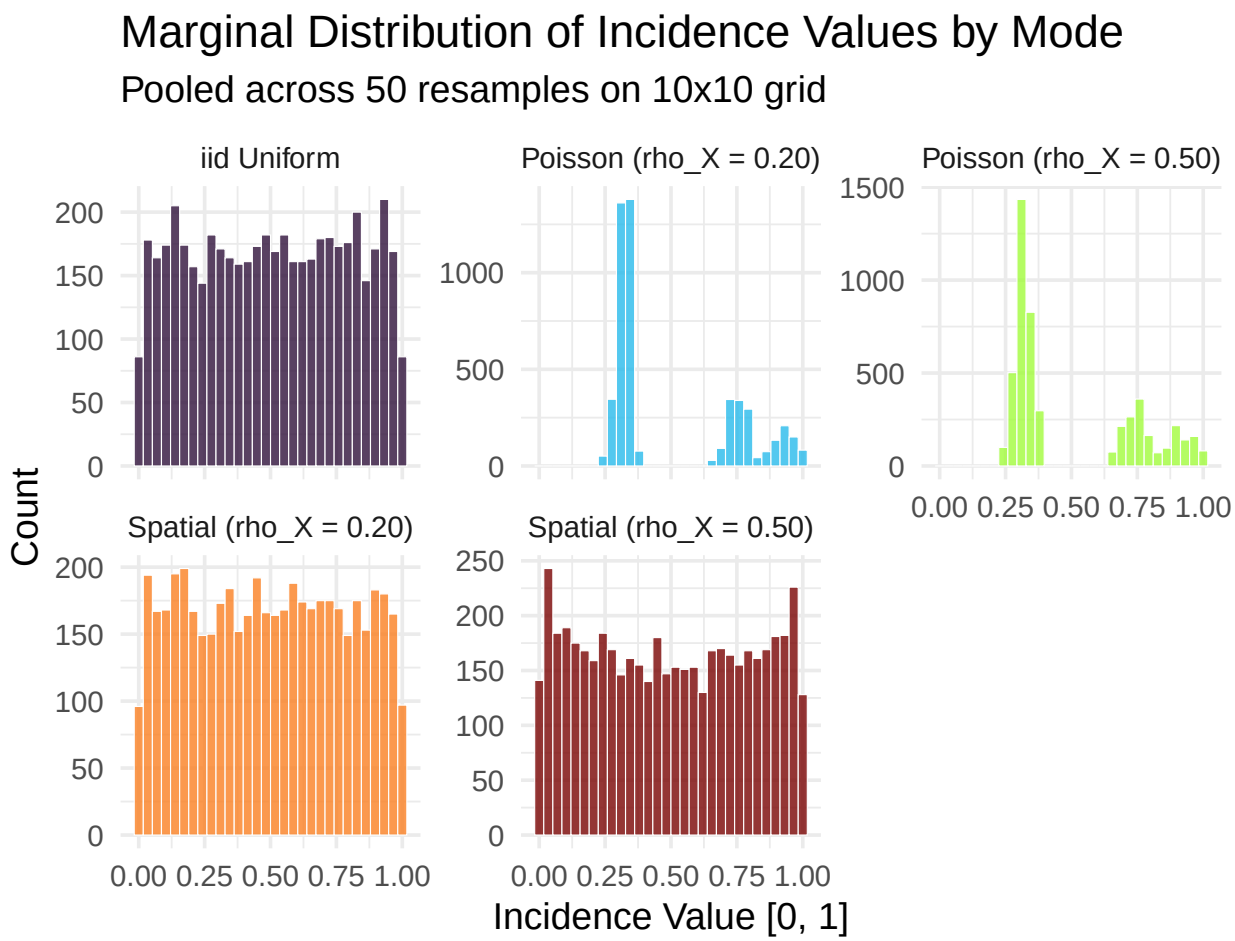

Figure 2: Marginal distribution of incidence values by mode, pooled across 50 resamples. All modes are marginally Uniform on [0,1]; spatial structure manifests in spatial covariance, not the marginal.

Table 2: 8 treatment assignment strategies evaluated in this simulation

ID	Name	Deterministic?
1	Design 1: Checkerboard	Yes
2	Design 2: High Incidence Focus	Yes
3	Design 3: Saturation Quadrants	No
4	Design 4: Isolation Buffer	No
5	Design 5: 2x2 Blocking	No
6	Design 6: Balanced Quartiles	No
7	Design 7: Balanced Halves	No
8	Design 8: Incidence-Guided Saturation Quadrants	No

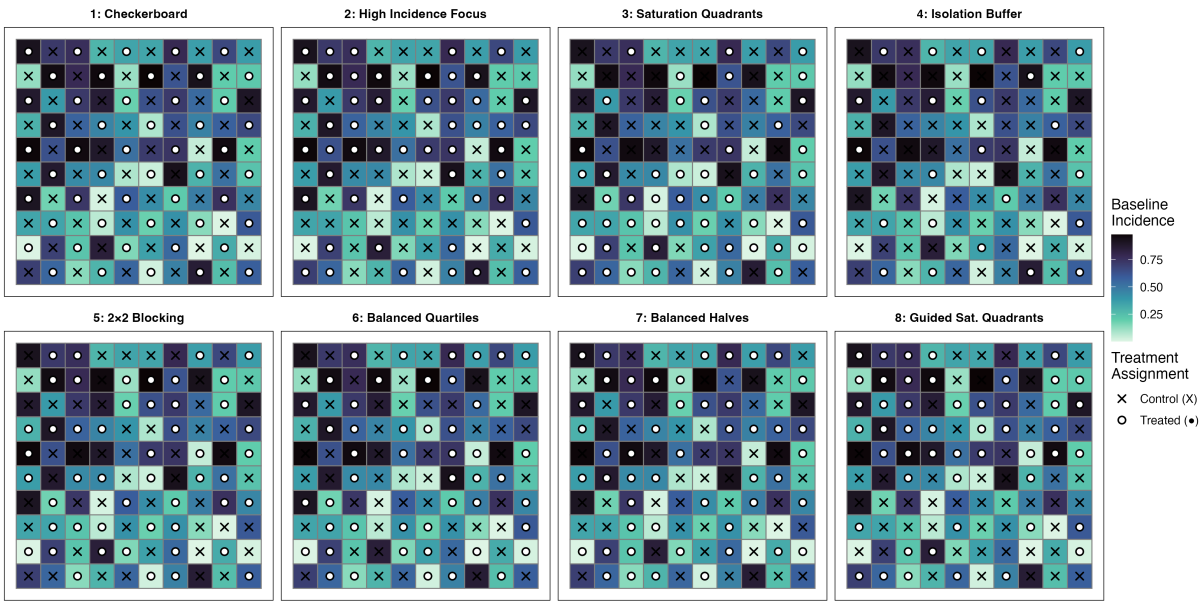

Figure 3: All 8 treatment assignment designs on a single 10×10 grid instance. Cell color represents baseline incidence (darker = higher burden). Circles ( ) = treated clusters; crosses (×) = control clusters.

5.1 Design 1: Checkerboard (Strict Spatial Stratification)

$$Z_i = (x_i + y_i) \mod 2$$

Treatment in an alternating checkerboard pattern. **Deterministic**, exactly 50% treated.

5.2 Design 2: High Incidence Focus

$$Z_i = \mathbf{1}[X_i > \text{median}(\mathbf{X})]$$

Treatment assigned to above-median incidence clusters. **Deterministic** given $\mathbf{X}$.

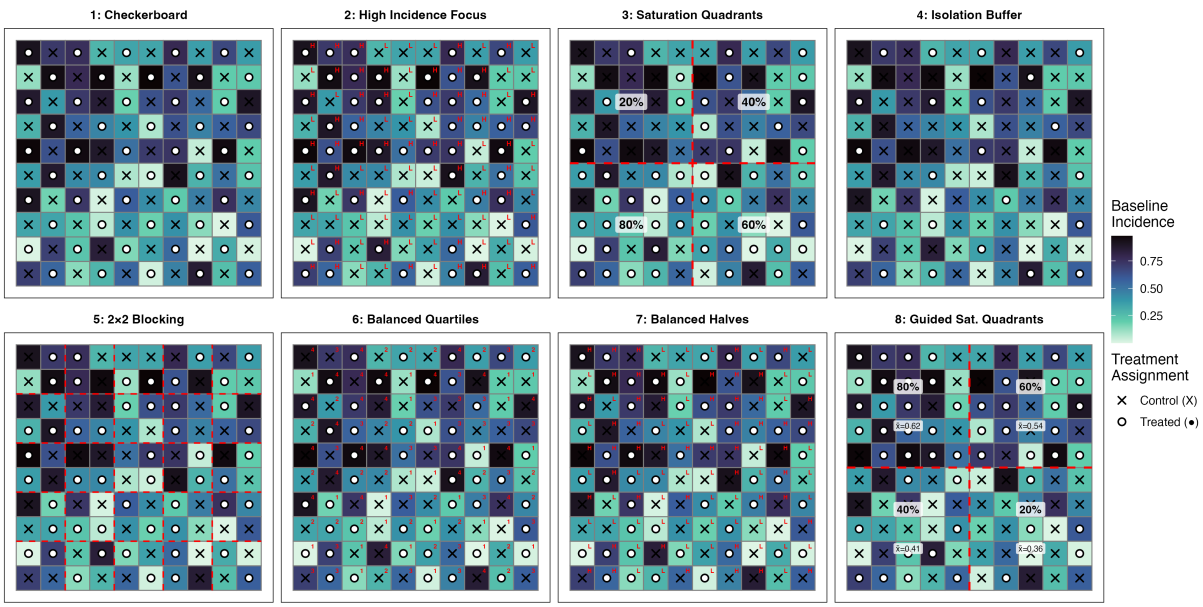

Figure 4: Same 8 designs with saturation percentage annotations. Percentage labels on the quadrant-based designs (Designs 3 and 8) show the within-quadrant treatment saturation rate assigned to each spatial region, illustrating how the two saturation quadrant strategies differ in how saturation levels are allocated.

5.3 Design 3: Saturation Quadrants

Grid partitioned into four spatial quadrants. Saturation levels $\{0.20, 0.40, 0.60, 0.80\}$ randomly assigned to quadrants. Within each quadrant, clusters randomly selected at the assigned rate.

5.4 Design 4: Isolation Buffer

Sequential treatment assignment: randomly select an available cluster, assign treatment, remove it and its neighbors from the available set. Treated clusters are never adjacent.

5.5 Design 5: 2x2 Blocking

Grid partitioned into non-overlapping 2×2 blocks. Within each block, exactly 2 of 4 clusters randomly assigned to treatment.

5.6 Design 6: Balanced Quartiles

Clusters stratified by incidence quartile. Within each stratum, approximately half randomly assigned to treatment.

5.7 Design 7: Balanced Halves

Clusters stratified into two groups by median baseline incidence. Within each half, approximately half randomly assigned to treatment. A 2-stratum analog of Design 6.

5.8 Design 8: Incidence-Guided Saturation Quadrants

Grid partitioned into four spatial quadrants (same as Design 3). Quadrants ranked by average baseline incidence and **deterministically** assigned saturation levels:

Rank (by $\bar{X}_{Q_k}$)	Saturation p_k
1 (highest)	0.80
2	0.60
3	0.40
4 (lowest)	0.20

Within each quadrant, clusters randomly selected at the assigned rate. Unlike Design 3, the saturation-to-quadrant mapping is guided by incidence data rather than random.

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6 Estimation Methods

6.1 Difference in Means (DIM)

The naive estimator:

$$\hat{\tau}_{\text{DIM}} = \bar{Y}_1 - \bar{Y}_0$$

DIM is biased for τ when spillover ($\gamma \neq 0$) or spatial dependence ($\rho \neq 0$) is present.

6.2 Maximum Likelihood Estimation (MLE)

The spatial autoregressive model is estimated via maximum likelihood using `spatialreg::lagsarlm()`:

$$Y_i = \rho \sum_j W_{ij} Y_j + \tau Z_i + \gamma \cdot \text{Spill}_i + \beta X_i + \varepsilon_i$$

MLE simultaneously estimates ρ and regression coefficients. The true spillover covariate Spill($\mathbf{Z}$) is included as an **oracle regressor**. Confidence intervals for τ use the asymptotic covariance matrix.

MLE is the primary estimator throughout this report. DIM serves only as a naive baseline.

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7 Simulation Design

7.1 Parameter Grid

Table 4: Simulation parameter grid (2,560 total scenarios)

Parameter	Values	Levels
Incidence mode	iid, poisson, spatial	5 configs (1 iid + 2 spatial + 2 poisson)
rho_X (incidence spatial autocorr.)	N/A (iid); 0.20, 0.50 (spatial, poisson)	—
nb_type (contiguity)	queen, rook	2
rho (outcome spatial autocorr.)	0, 0.01, 0.2, 0.5	4
gamma (spillover magnitude)	0.5, 0.6, 0.7, 0.8	4
spill_type	both, control_only	2
design_id	1, 2, 3, 4, 5, 6, 7, 8	8
TOTAL		2,560

7.2 Resampling Structure

Mode	Design resamples (R_d)	Outcome resamples (R_o)	Iterations / scenario
MLE	25	10	250

7.3 Per-Scenario Reproducibility

Each scenario uses a deterministic seed:

$$\text{seed}_s = \texttt{digest::digest2int}(\texttt{paste}(\text{inc_mode}, \rho_X, \text{nb_type}, \rho, \gamma, \text{spill_type}, d_{\text{id}}, \text{sep} = \text{"|"}))$$

This ensures results are reproducible and order-independent.

---

8 Evaluation Metrics

For each scenario, let $\hat{\tau}_1, \ldots, \hat{\tau}_K$ denote the K estimates of τ (excluding convergence failures).

- **Bias**: Bias $= \bar{\hat{\tau}} - \tau$
- **Standard Deviation**: SD $= \sqrt{\frac{1}{K-1} \sum_{k=1}^{K} \left(\hat{\tau}_k - \bar{\hat{\tau}}\right)^2}$

- **Mean Squared Error**: $\text{MSE} = \text{Bias}^2 + \text{SD}^2$ — **primary ranking criterion**
- **Coverage Probability**: $\text{Coverage} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{1}\left[\tau \in \left(\hat{\tau}_{k,L},\ \hat{\tau}_{k,U}\right)\right]$
- **Convergence Failure Rate**: Proportion of iterations where MLE failed to converge

---

9 Results: Overall Performance

9.1 Overall Summary

Table 6: MLE: Summary statistics across all 2,560 scenarios

Metric	Min	Mean	Max
Bias	-0.965	-0.026	0.811
MSE	0.012	0.210	4.812
Coverage	0.000	0.890	1.000
Fail Rate	NA	0.000	NA

9.2 Design Comparison by MSE

Table 7: MLE: Design performance averaged across all scenarios and incidence modes

Design	Mean Bias	Median MSE	Mean MSE	Mean Coverage	Fail Rate
Design 8	0.025	0.059	0.079	0.941	0
Design 3	0.016	0.065	0.080	0.939	0
Design 6	0.010	0.076	0.102	0.938	0
Design 7	0.012	0.079	0.110	0.937	0
Design 2	0.013	0.086	0.133	0.966	0
Design 4	0.012	0.146	0.148	0.939	0
Design 5	0.007	0.120	0.228	0.937	0
Design 1	-0.305	0.456	0.802	0.527	0

9.3 MSE by Design and Neighbor Type

9.4 Coverage by Design

9.5 Bias-Variance Decomposition

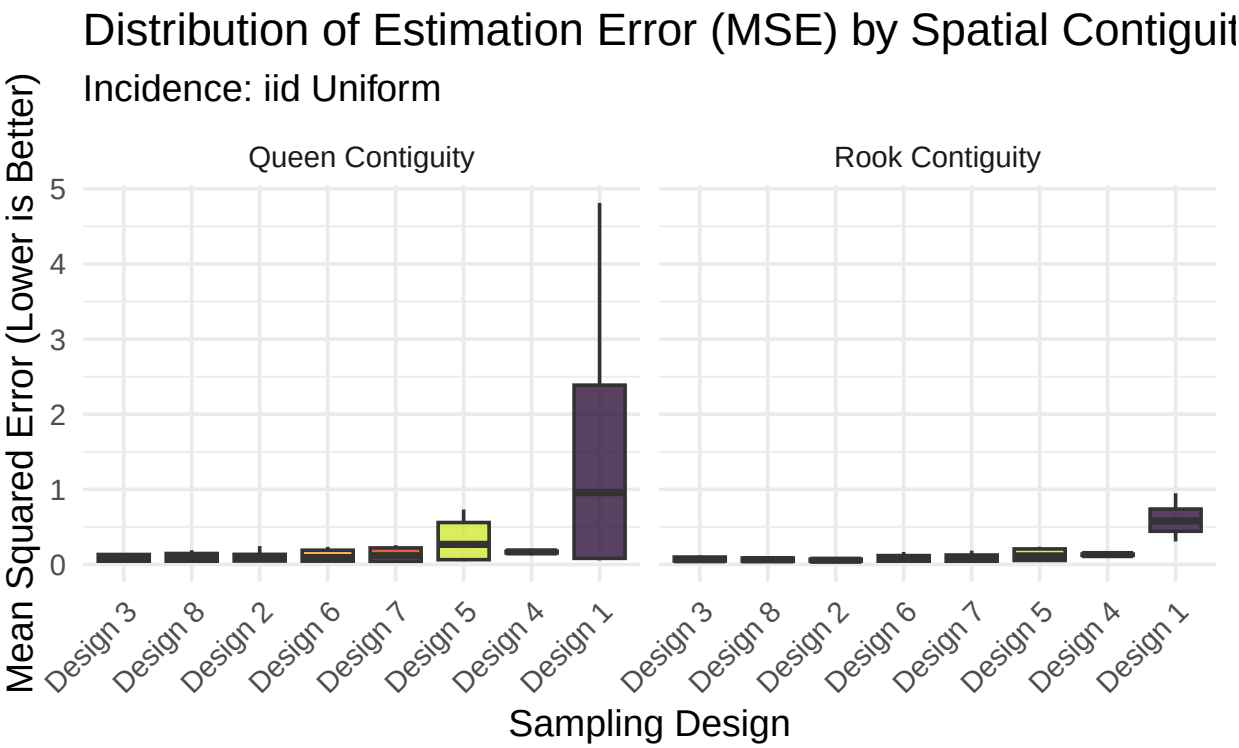

Figure 5: MLE: Distribution of MSE by design, faceted by spatial contiguity structure.

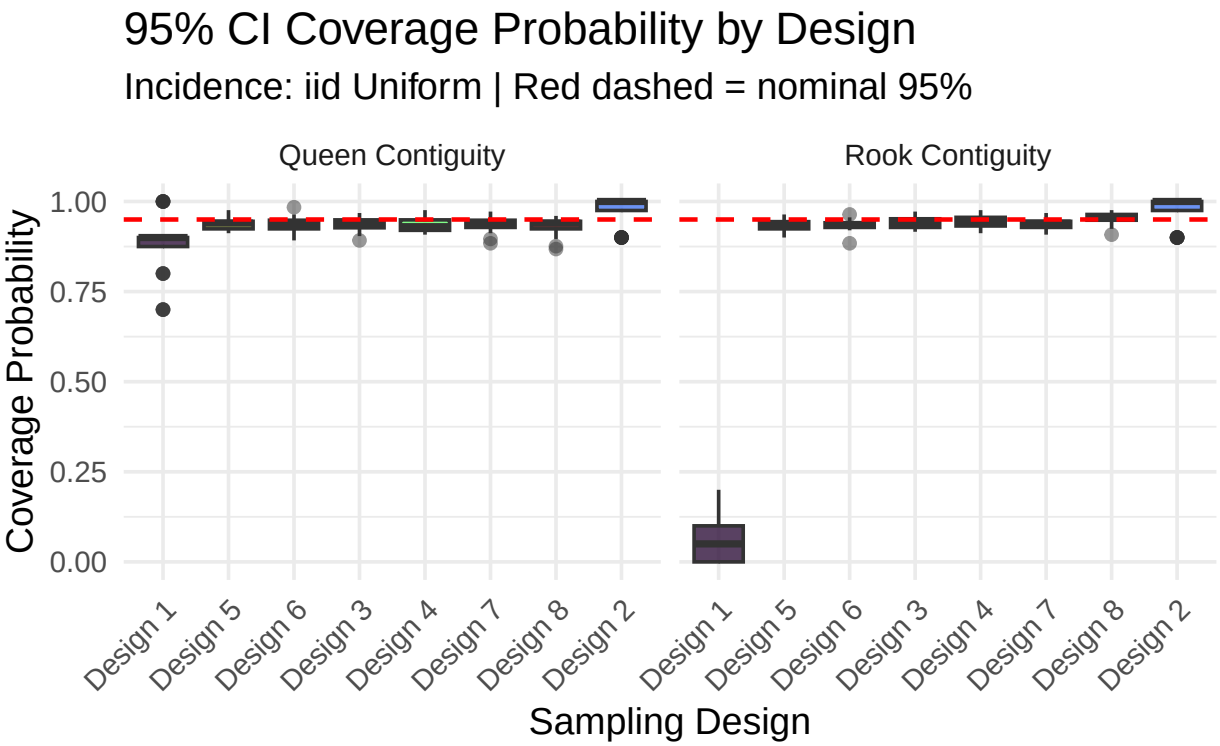

Figure 6: MLE: Coverage probability distribution by design. Red dashed line = nominal 95% target.

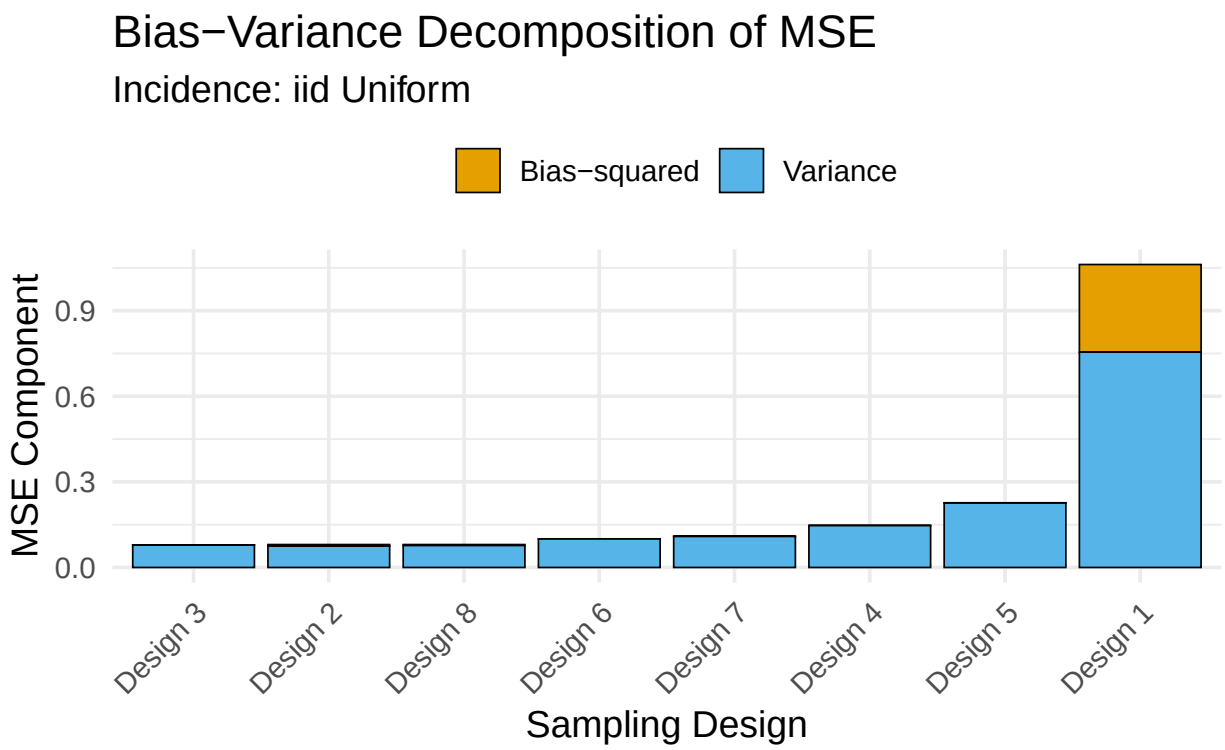

Figure 7: MLE: Bias-squared (dark) and variance (light) components of MSE per design for iid Uniform incidence.

10 Results: Formal Statistical Comparisons

The descriptive summaries above rank designs by mean MSE, but do not establish whether observed differences are *statistically significant*. Here we apply non-parametric hypothesis tests to the paired MSE values across the 320 unique parameter blocks. Each block is a unique combination of simulation parameters (Incidence Mode × Neighbor Type × ρ × γ × Spillover Type); within each block all 8 designs are evaluated under identical conditions, creating a paired repeated-measures structure.

Three complementary tests are used: (1) the **Friedman test** (omnibus — is there any significant difference among designs?), (2) **Nemenyi post-hoc** comparisons with family-wise error control (which specific pairs differ?), and (3) pairwise **Wilcoxon signed-rank tests** with Holm correction (sensitivity check). Significance level $\alpha = 0.05$ throughout.

10.1 Friedman Omnibus Test

Table 8: Friedman omnibus test: are any design MSE ranks significantly different?

Chi-squared	df	P-value	N Blocks	Decision
1091.90	7	<2e-16	320	Reject H0

The Friedman test ($\chi^2 = 1091.90$, $p < 2.2 \times 10^{-16}$) decisively rejects the null hypothesis that all 8 designs perform equally. The table below shows each design's **average MSE rank** across all blocks (rank 1 = best):

Table 9: Average MSE rank per design across all 320 blocks (1 = best).

Design	Average Rank
Design 8	**2.666**
Design 3	**2.709**
Design 6	3.622
Design 2	3.819
Design 7	3.909
Design 4	5.772
Design 5	6.222
Design 1	7.281

10.2 Critical Difference Diagram

How to read this diagram: Designs positioned closer together are more similar in performance. A thick bar connecting two designs means their difference is *not* statistically significant. The red arrow labelled "CD" shows the critical difference threshold — any two designs separated by more than this distance are significantly different.

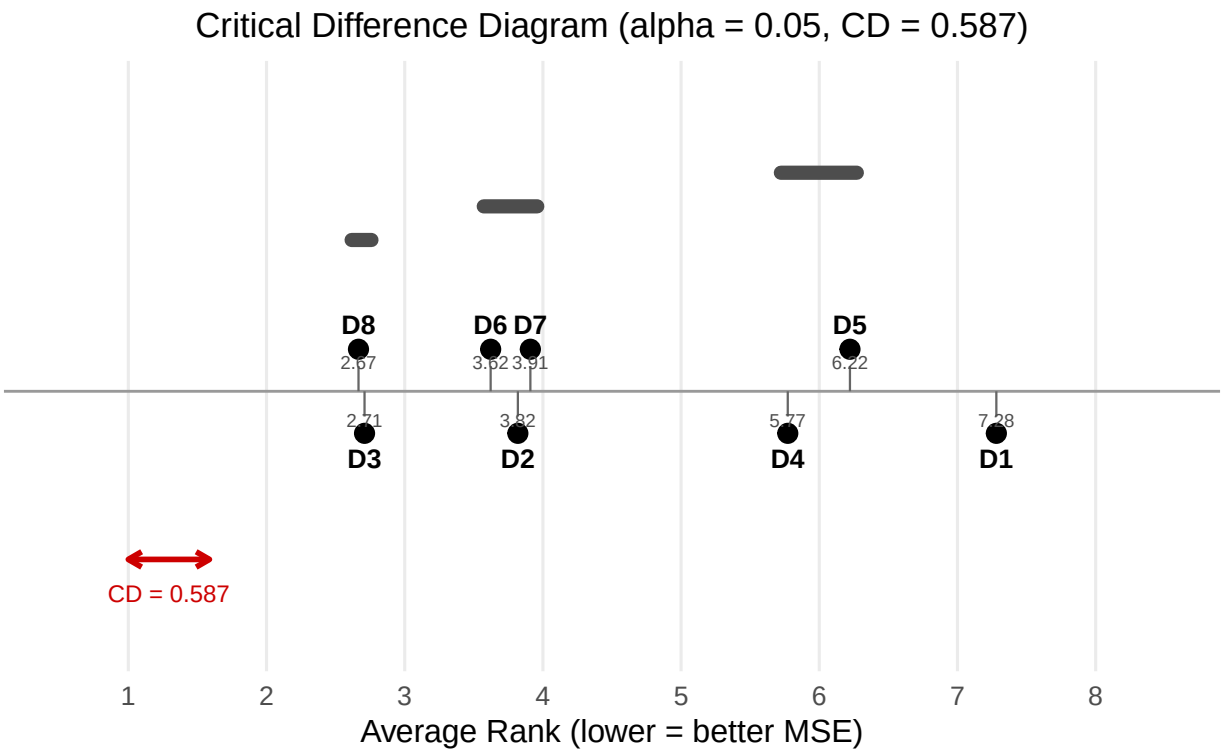

Figure 8: Critical Difference (CD) diagram from Nemenyi post-hoc analysis. Designs are placed on a number line by average MSE rank (lower = better). Thick bars connect designs that are NOT significantly different at alpha = 0.05. The red arrow shows the CD threshold.

The headline finding: **Design 3 (Saturation Quadrants) and Design 8 (Incidence-Guided Saturation Quadrants) form the top equivalence group** — statistically indistinguishable from each other, and significantly better than all other designs. **Design 1 (Checkerboard) is significantly worse than all alternatives** (Nemenyi $p < 0.001$ vs. every other design).

10.3 Pairwise Significance: MSE Distributions

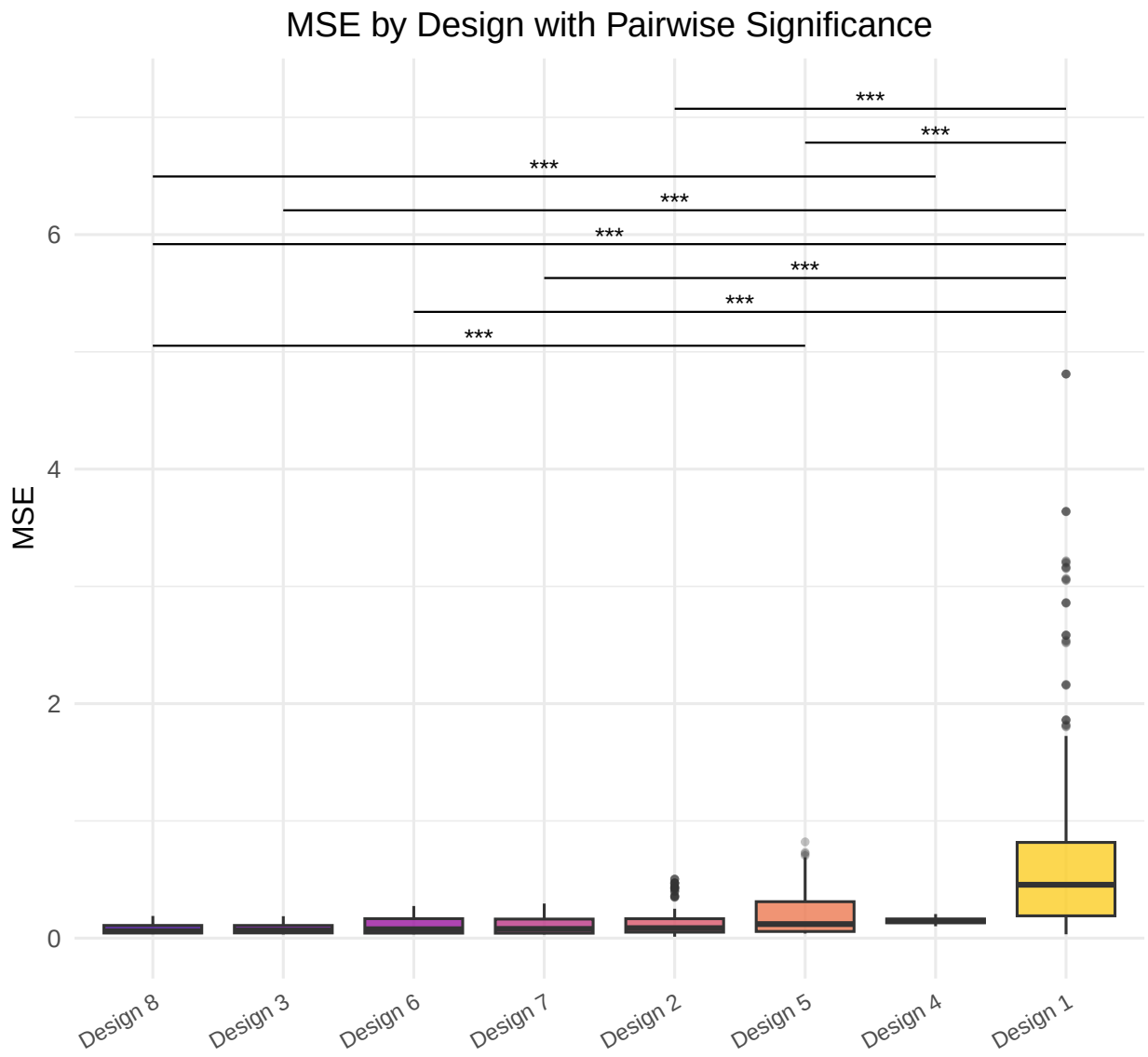

Figure 9: MSE distribution by design (ordered by median MSE, best on left). Significance brackets show pairwise Wilcoxon signed-rank test results (Holm-adjusted). Stars: *** p < 0.001, ** p < 0.01, * p < 0.05.

The boxplot confirms the rank-based findings: Designs 3 and 8 have the lowest and most compact MSE distributions, while Design 1 has both the highest median and the widest spread (high variance alongside high bias). Significance brackets annotate the most informative pairwise comparisons.

10.4 Pairwise P-value Heatmap

Figure 10: Pairwise Wilcoxon signed-rank p-values (Holm-adjusted). Red = strongly significant ($p < 0.001$), orange = significant ($p < 0.01$ or < 0.05), grey/green = not significant.

Of the 28 pairwise comparisons, **24 are statistically significant** under Wilcoxon (Holm) and **23** under Nemenyi post-hoc. The 4 non-significant pair(s) are:

Table 10: Design pairs with no statistically significant MSE difference.

Design A	Design B	Wilcoxon p	Nemenyi p
Design 2	Design 6	0.4418	0.9721
Design 2	Design 7	0.8893	0.9998
Design 3	Design 8	0.2467	1.0000

Design 4	Design 5	0.1194	0.2804

10.5 Robustness: Conditional Tests by Parameter

To confirm that the design rankings are not an artifact of any particular parameter setting, we repeat the Friedman test within strata defined by each simulation parameter individually.

Table 11: Friedman test within each parameter stratum. 'Best Group' lists designs statistically indistinguishable from the best-ranked design.

Parameter	Stratum	Chi-sq	P-value	N Blocks	Best Group
Rho	Rho = 0	336.74	< 2.2e-16	80	D3, D2, D8, D6
Rho	Rho = 0.01	272.14	< 2.2e-16	80	D3, D8, D6, D2
Rho	Rho = 0.2	297.57	< 2.2e-16	80	D8, D3, D6
Rho	Rho = 0.5	223.59	< 2.2e-16	80	D8, D3
Gamma	Gamma = 0.5	266.69	< 2.2e-16	80	D3, D8, D6, D2
Gamma	Gamma = 0.6	272.65	< 2.2e-16	80	D8, D3, D6
Gamma	Gamma = 0.7	279.50	< 2.2e-16	80	D8, D3, D6, D2
Gamma	Gamma = 0.8	274.22	< 2.2e-16	80	D3, D8, D6, D2
Spillover_Type	Spillover_Type = both	597.10	< 2.2e-16	160	D7, D6, D8, D3
Spillover_Type	Spillover_Type = control_only	846.04	< 2.2e-16	160	D8, D3
Neighbor_Type	Neighbor_Type = queen	475.39	< 2.2e-16	160	D3, D8, D2
Neighbor_Type	Neighbor_Type = rook	658.35	< 2.2e-16	160	D8, D3, D6
Incidence_Mode	Incidence_Mode = iid	290.56	< 2.2e-16	64	D8, D2, D3
Incidence_Mode	Incidence_Mode = poisson	390.80	< 2.2e-16	128	D8, D3, D6
Incidence_Mode	Incidence_Mode = spatial	488.47	< 2.2e-16	128	D3, D8, D2

The Friedman test is **significant in every stratum** ($p < 2.2 \times 10^{-16}$ in all cases). Design 3 and Design 8 appear in the top equivalence group across all parameter levels, confirming that the recommendation is robust.

> For full methods background (Friedman test derivation, CD formula, Wilcoxon procedure with references) and complete pairwise p-value matrices, see the companion report: **Statistical Comparison of Treatment Assignment Designs** (`11_statistical_comparisons_report.pdf` in the results directory).

10.6 Summary of Statistical Findings

Taken together, the Friedman test, Nemenyi post-hoc comparisons, and pairwise Wilcoxon signed-rank tests converge on a clear and consistent picture:

- **Design differences are real, not chance.** The Friedman omnibus test ($\chi^2 = 1091.90$, $p < 2.2 \times 10^{-16}$) provides overwhelming evidence that the 8 designs do not perform equally. The average rank gap between best and worst is 4.61 points on an 8-point scale.

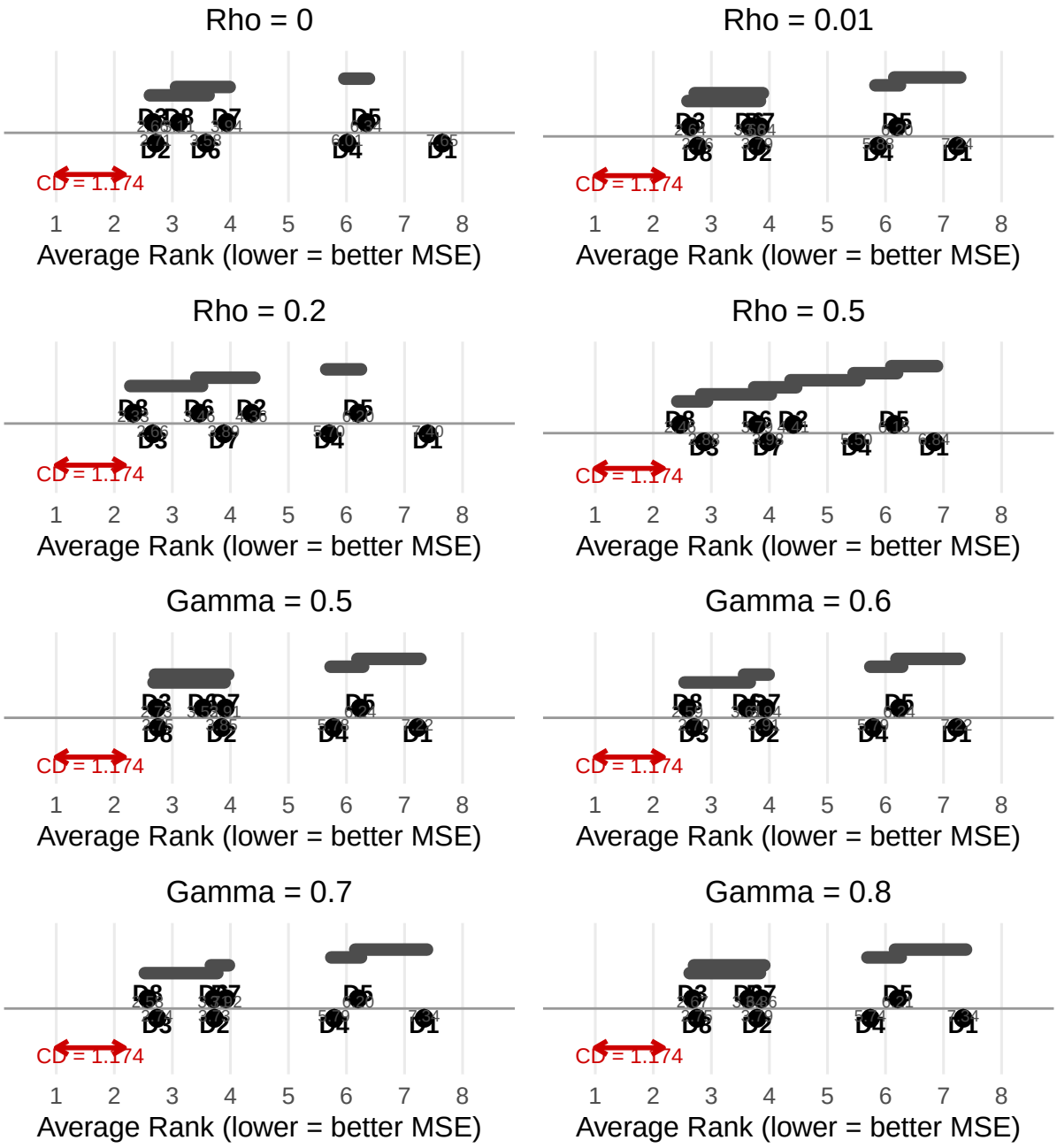

Figure 11: CD diagrams by outcome spatial autocorrelation (Rho, top) and spillover magnitude (Gamma, bottom). Rankings are consistent across levels.

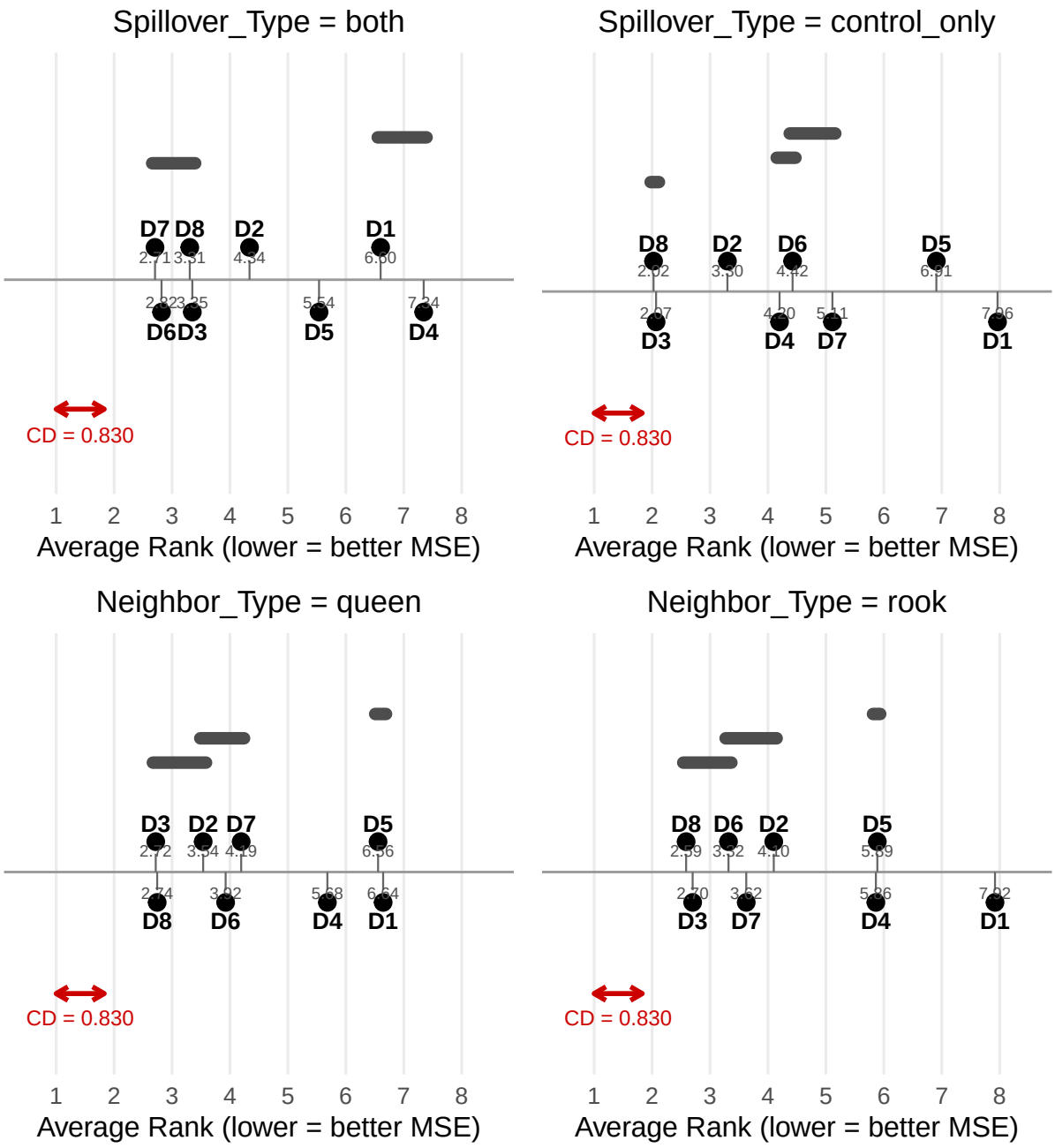

Figure 12: CD diagrams by spillover type (top) and neighbor contiguity type (bottom). Design rankings are stable regardless of spillover structure or contiguity definition.

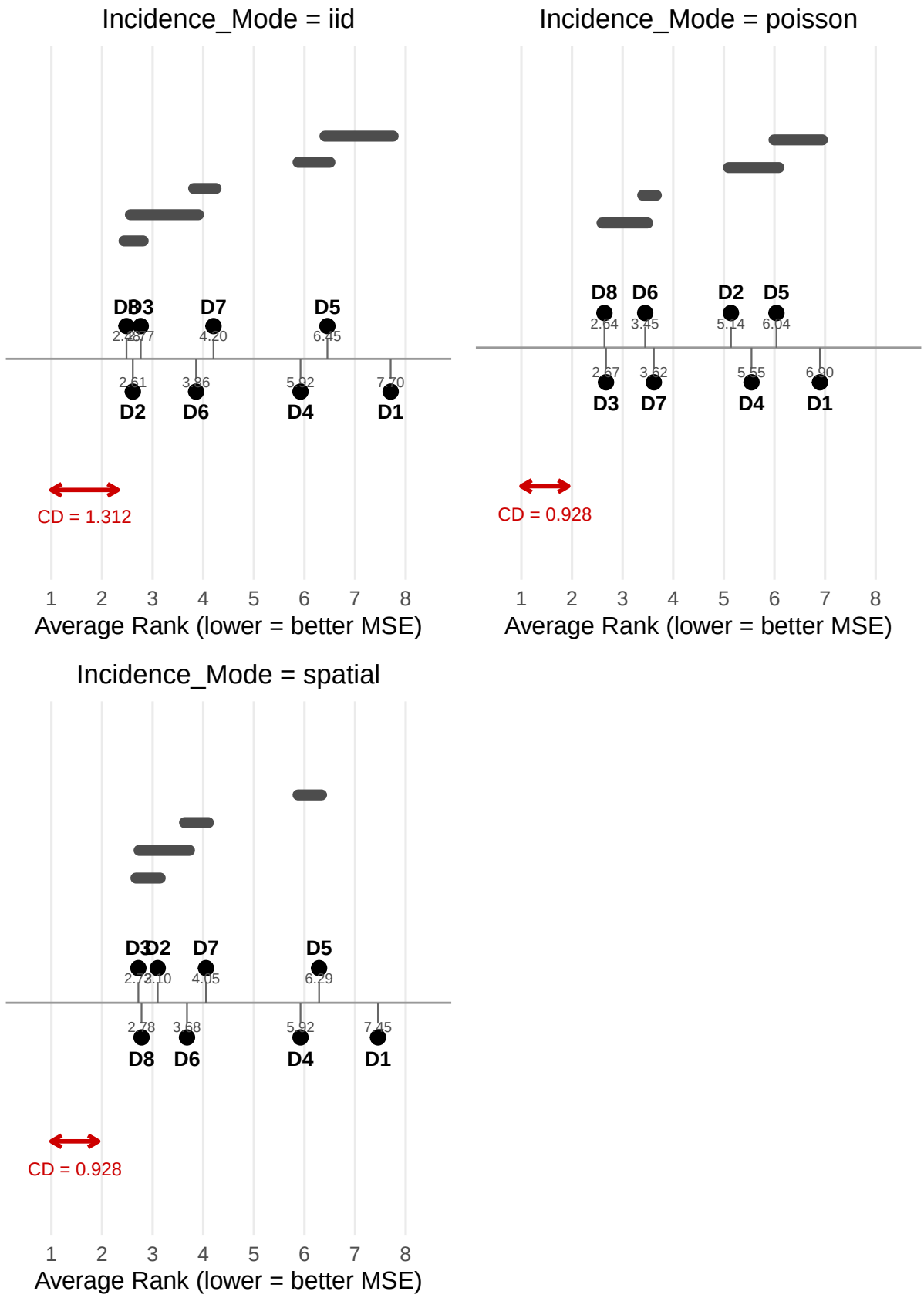

Figure 13: CD diagrams by incidence generation mode. Design rankings hold across all five incidence configurations.

- **Design 3 and Design 8 are the recommended designs.** Saturation Quadrants (Design 8, avg rank 2.67) and Incidence-Guided Saturation Quadrants (Design 3, avg rank 2.71) form the top equivalence group — they are statistically indistinguishable from each other and significantly better than every other design. When prior incidence data are available, Design 8 may offer a marginal advantage; otherwise Design 3 is equally well-supported.
- **Avoid Design 1 (Checkerboard).** With an average rank of 7.28, the Checkerboard pattern is significantly worse than all alternatives under MLE estimation. Its poor performance is consistent across all parameter settings.
- **23 of 28 design pairs are significantly different** (Nemenyi post-hoc; 24 under Wilcoxon/Holm). The few non-significant pairs involve mid-tier designs with similarly modest MSE profiles.
- **Rankings are robust.** The Friedman test is significant in every parameter stratum tested ($p < 2.2 \times 10^{-16}$ across all levels of ρ, γ, spillover type, neighbor type, and incidence mode). The top equivalence group is stable regardless of the specific data-generating conditions, giving investigators confidence that the design recommendation generalises beyond any single scenario.

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11 Results: Design Rankings by Incidence Mode (Q1)

11.1 Cross-Config MSE Rankings

11.2 Cross-Config Coverage Rankings

11.3 Detailed Rankings by Incidence Config

Table 12: MLE design rankings for iid Uniform

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D3: Sat. Quadrants**	**0.0788**	**0.940**	**0.0108**
2	D2: High Inc. Focus	0.0800	0.975	0.0124
3	D8: Inc.-Guided Sat. Q.	0.0800	0.942	0.0116
4	D6: Balanced Quartiles	0.1000	0.935	0.0058
5	D7: Balanced Halves	0.1093	0.937	0.0109
6	D4: Isolation Buffer	0.1468	0.940	-0.0053
7	D5: 2x2 Blocking	0.2258	0.936	-0.0047
8	D1: Checkerboard	1.0586	0.469	-0.3268

Table 13: MLE design rankings for Poisson (rho X = 0.20)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D8: Inc.-Guided Sat. Q.**	**0.0778**	**0.939**	**0.0488**

2	D3: Sat. Quadrants	0.0841	0.929	0.0329
3	D6: Balanced Quartiles	0.1076	0.936	0.0240
4	D7: Balanced Halves	0.1114	0.936	0.0192
5	D4: Isolation Buffer	0.1472	0.939	0.0164
6	D2: High Inc. Focus	0.2017	0.956	0.0095
7	D5: 2x2 Blocking	0.2315	0.935	0.0251
8	D1: Checkerboard	0.9211	0.512	-0.2190

Table 14: MLE design rankings for Poisson (rho X = 0.50)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D3: Sat. Quadrants**	**0.0790**	**0.939**	**0.0112**
2	D8: Inc.-Guided Sat. Q.	0.0811	0.938	0.0192
3	D6: Balanced Quartiles	0.1024	0.942	0.0067
4	D7: Balanced Halves	0.1045	0.941	0.0113
5	D4: Isolation Buffer	0.1447	0.940	0.0294
6	D2: High Inc. Focus	0.2072	0.966	-0.0118
7	D5: 2x2 Blocking	0.2300	0.938	0.0138
8	D1: Checkerboard	0.6944	0.556	-0.3206

Table 15: MLE design rankings for Spatial (rho X = 0.20)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D8: Inc.-Guided Sat. Q.**	**0.0778**	**0.948**	**0.0351**
2	D3: Sat. Quadrants	0.0810	0.940	0.0140
3	D2: High Inc. Focus	0.0982	0.963	0.0483
4	D6: Balanced Quartiles	0.1007	0.942	0.0036
5	D7: Balanced Halves	0.1129	0.938	0.0117
6	D4: Isolation Buffer	0.1504	0.940	0.0113
7	D5: 2x2 Blocking	0.2291	0.938	-0.0024
8	D1: Checkerboard	0.7911	0.522	-0.3468

Table 16: MLE design rankings for Spatial (rho X = 0.50)

Rank	Design	Avg MSE	Avg Coverage	Avg Bias
1	**D3: Sat. Quadrants**	**0.0754**	**0.945**	**0.0101**
2	D8: Inc.-Guided Sat. Q.	0.0774	0.937	0.0086
3	D2: High Inc. Focus	0.0781	0.970	0.0087
4	D6: Balanced Quartiles	0.1012	0.936	0.0091
5	D7: Balanced Halves	0.1096	0.935	0.0076
6	D4: Isolation Buffer	0.1498	0.936	0.0060
7	D5: 2x2 Blocking	0.2221	0.940	0.0028
8	D1: Checkerboard	0.5461	0.575	-0.3112

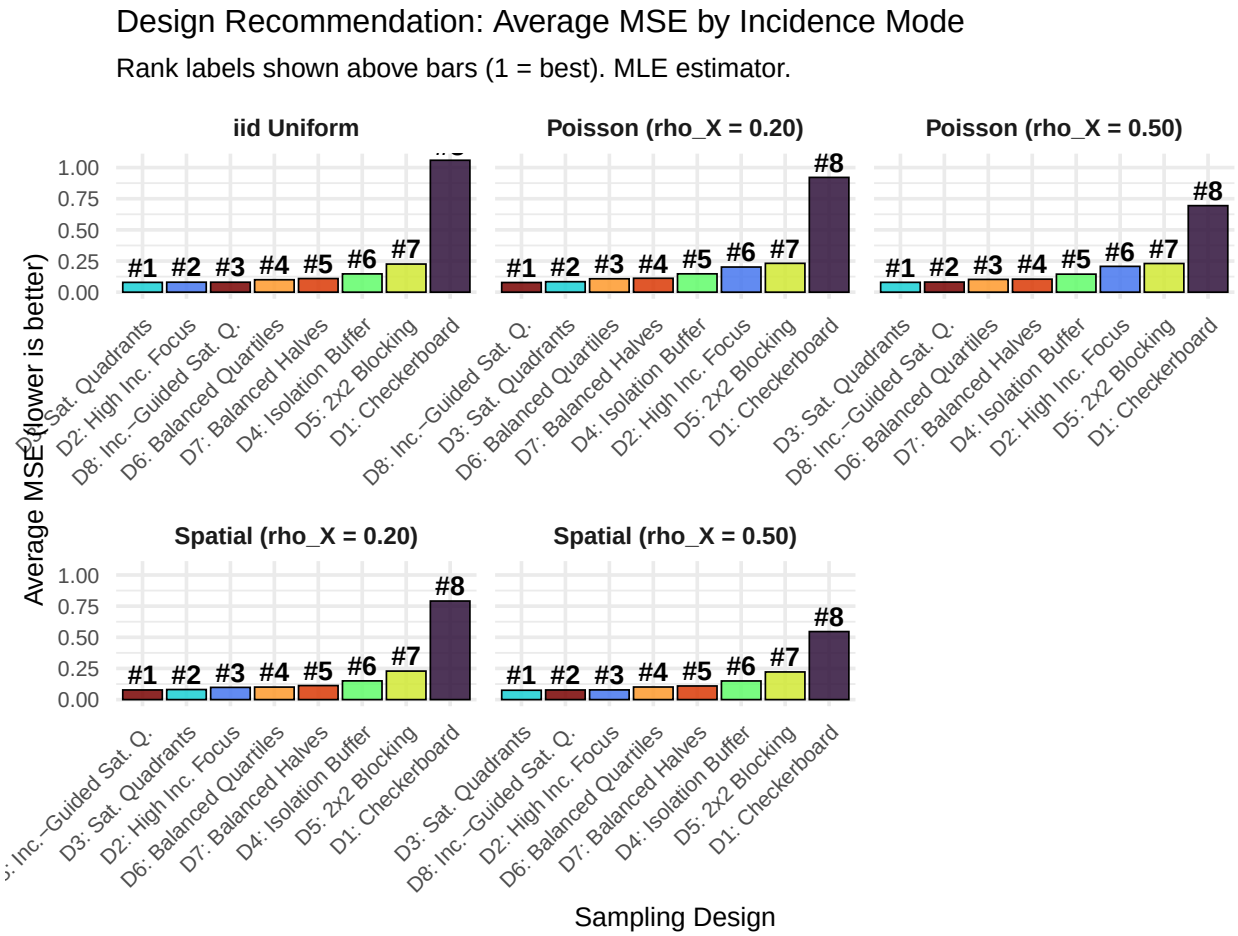

Figure 14: Average MSE per design for each incidence configuration. Bars sorted left-to-right (lowest = best). Rank labels above bars.

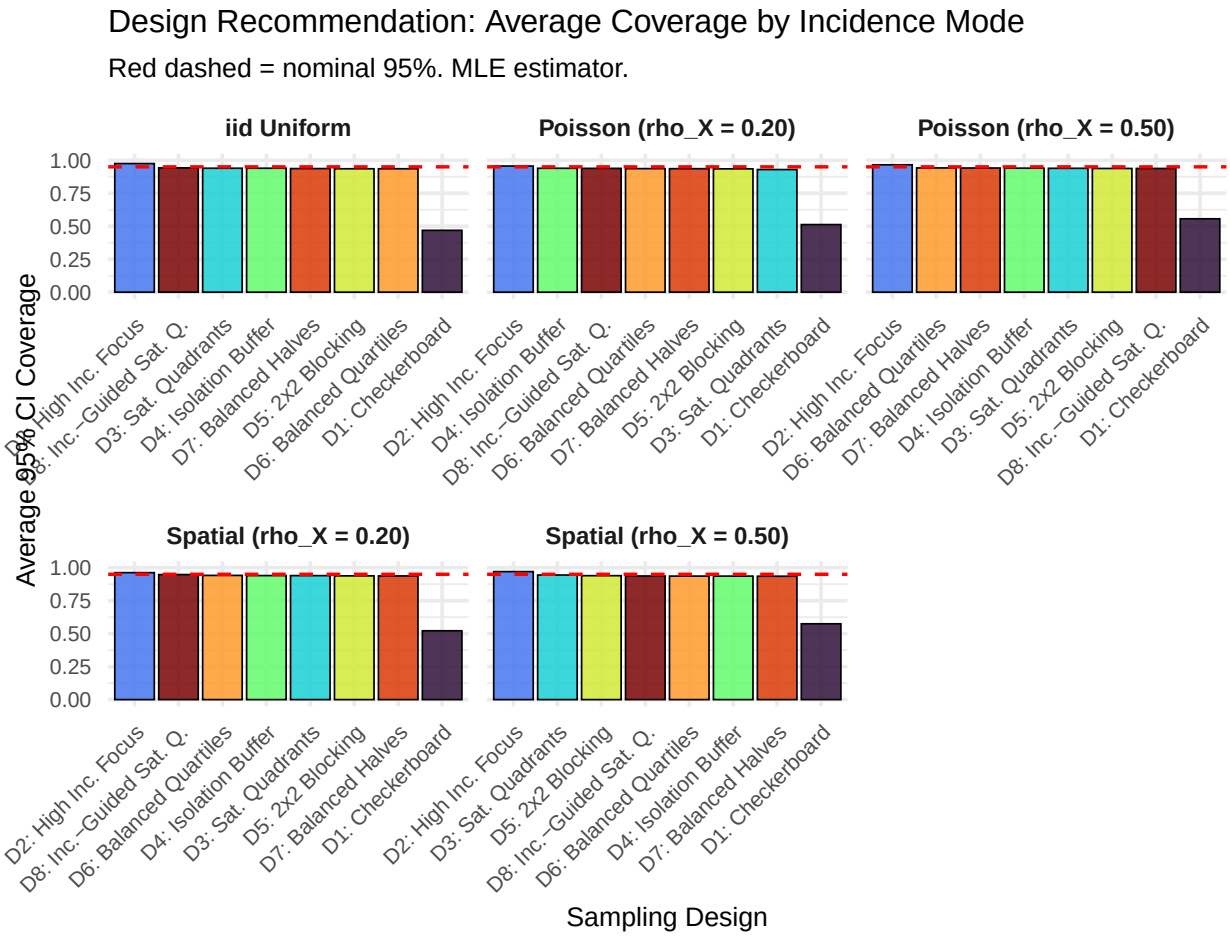

Figure 15: Average 95% CI coverage probability per design for each incidence configuration. Red dashed line = nominal 0.95.

12 Results: Parameter Sensitivity (Q2)

12.1 Effect of Outcome Spatial Autocorrelation (Rho)

12.2 Effect of Spillover Magnitude (Gamma)

12.3 Sensitivity Analysis (Eta-squared)

Table 17: MLE: Eta-squared (proportion of MSE variance explained) by parameter, for iid Uniform incidence.

Design	Rho	Gamma	Spillover_Type	Neighbor_Type
Design 1	0.0479	0.0043	0.3231	0.1334
Design 2	0.0653	0.0000	0.4654	0.1680
Design 3	0.0181	0.0007	0.8413	0.0451
Design 4	0.0202	0.0059	0.0003	0.5531
Design 5	0.0008	0.0010	0.6135	0.1901
Design 6	0.0084	0.0033	0.7946	0.0765
Design 7	0.0030	0.0012	0.7912	0.0858
Design 8	0.0225	0.0016	0.7064	0.1122

12.4 MSE Per Design Across Gamma

```
$`Design 1`
```

```
$`Design 2`
```

```
$`Design 3`
```

```
$`Design 4`
```

```
$`Design 5`
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```
$`Design 6`
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```
$`Design 7`
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$`Design 8`
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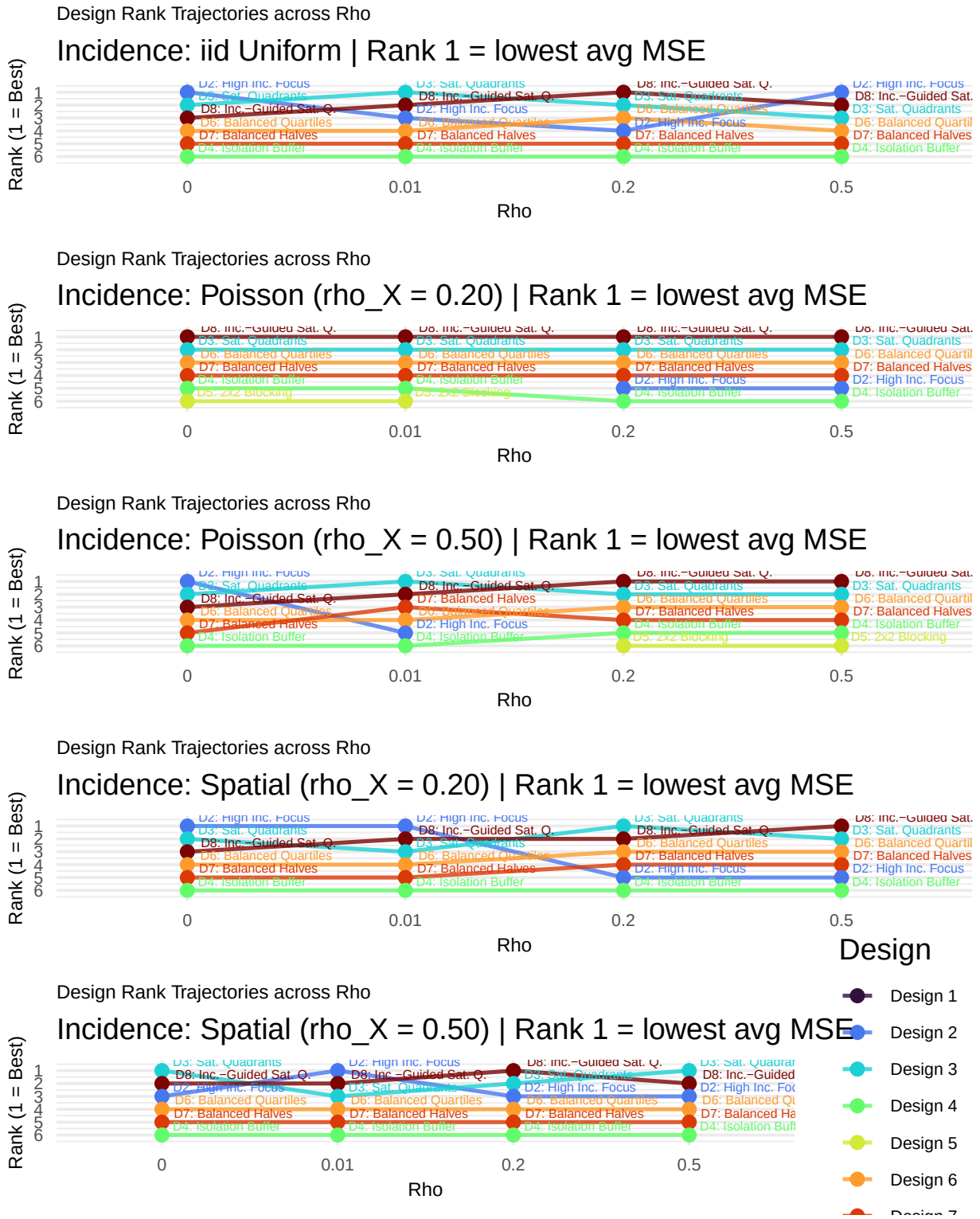

Figure 16: Design rank trajectories across Rho for each incidence config. Rank 1 = best. Flat lines indicate robust designs; crossing lines reveal rank reversals.

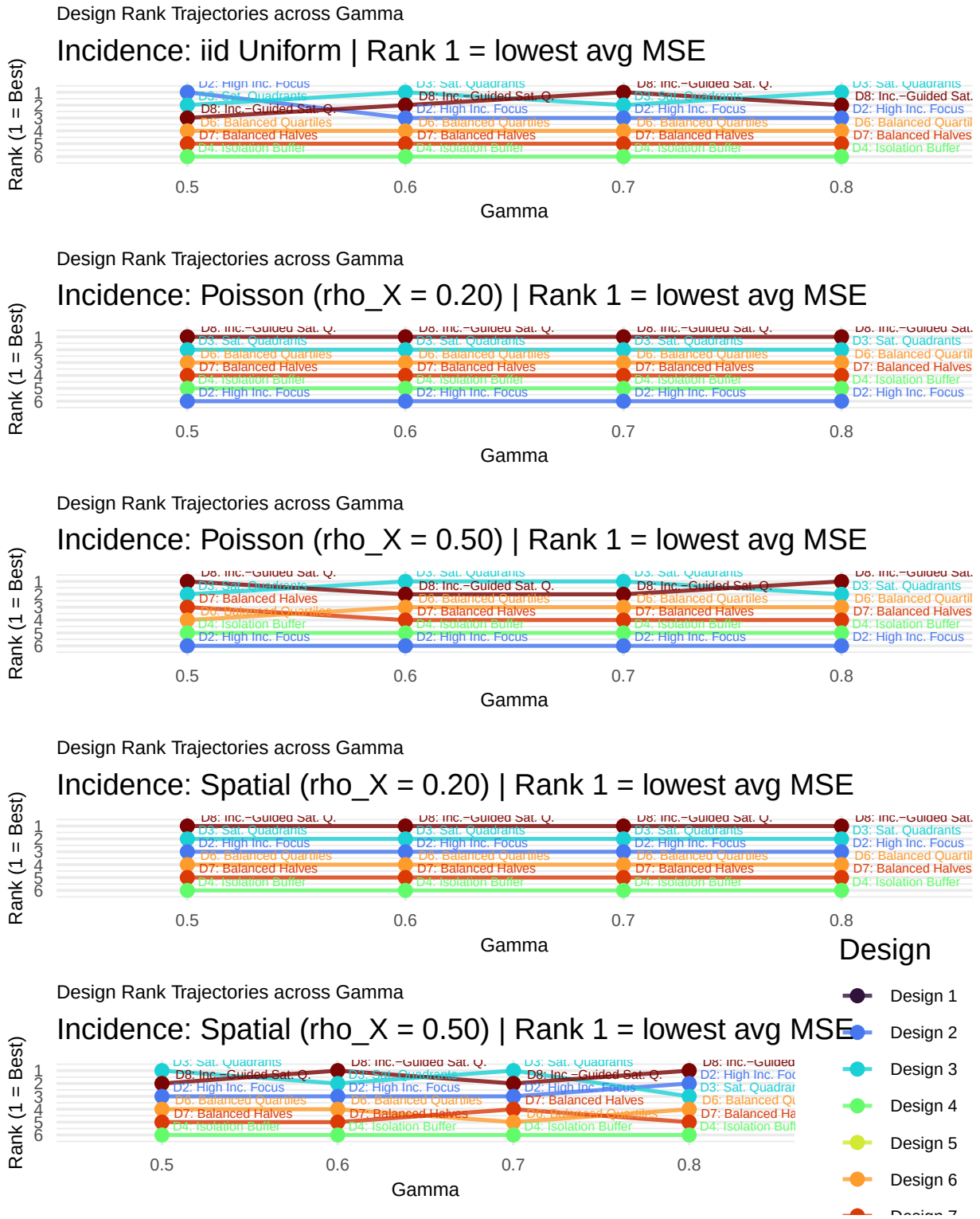

Figure 17: Design rank trajectories across Gamma. Gamma has minimal effect on design rankings under MLE: all trajectory lines are nearly flat.

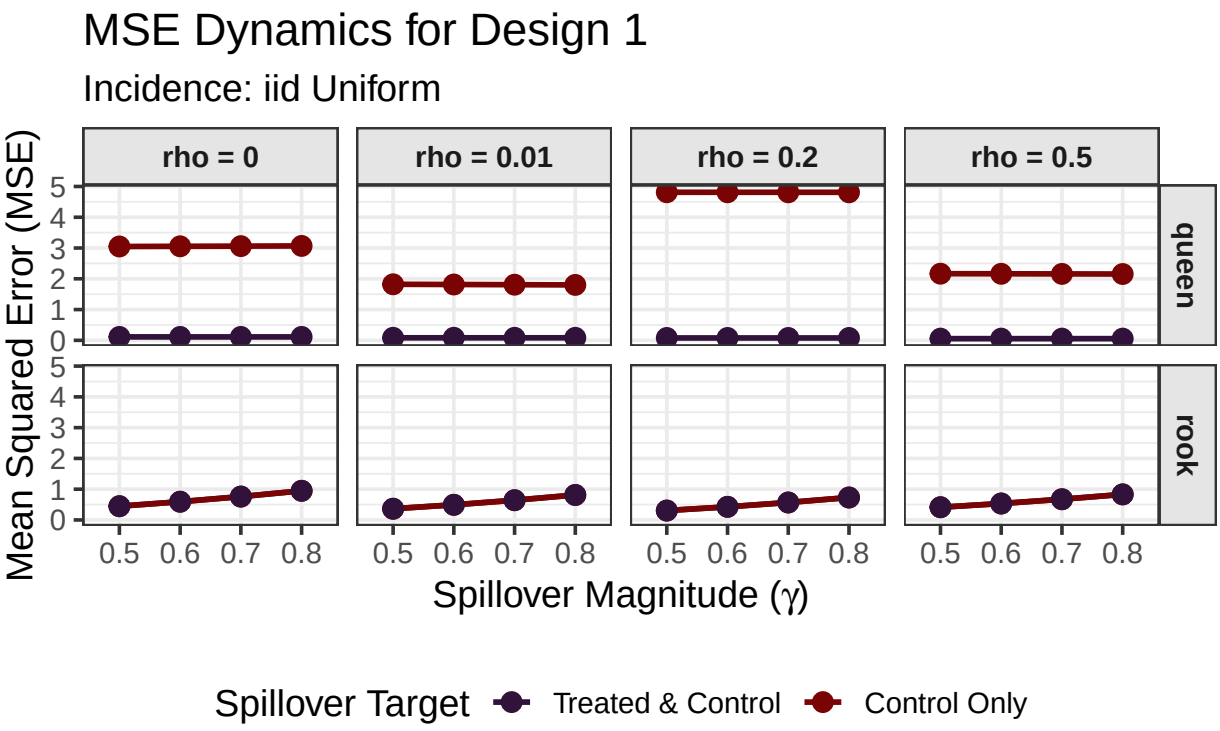

Figure 18: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

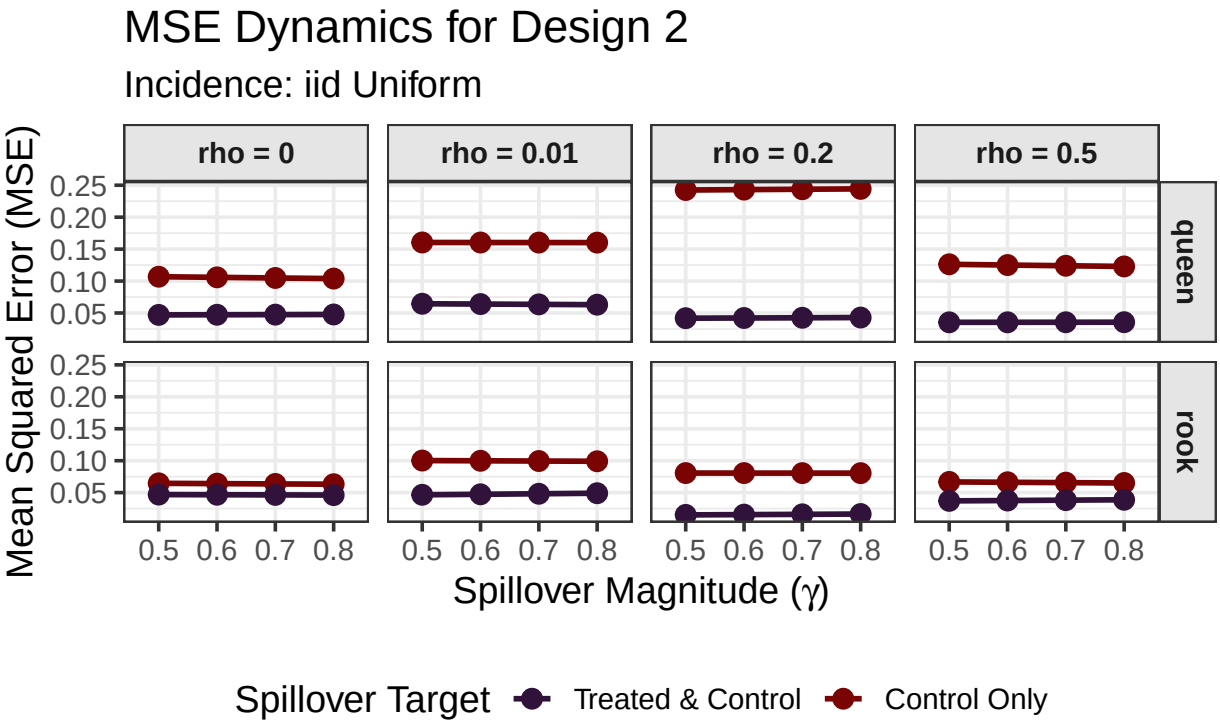

Figure 19: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

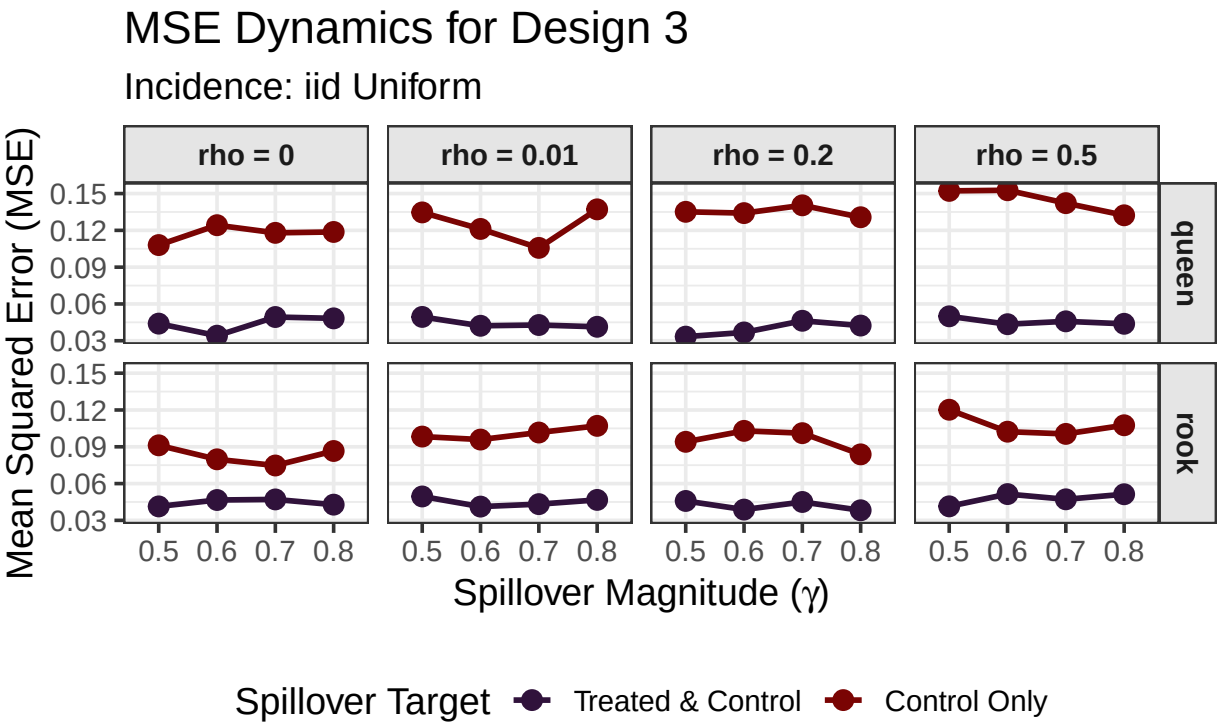

Figure 20: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

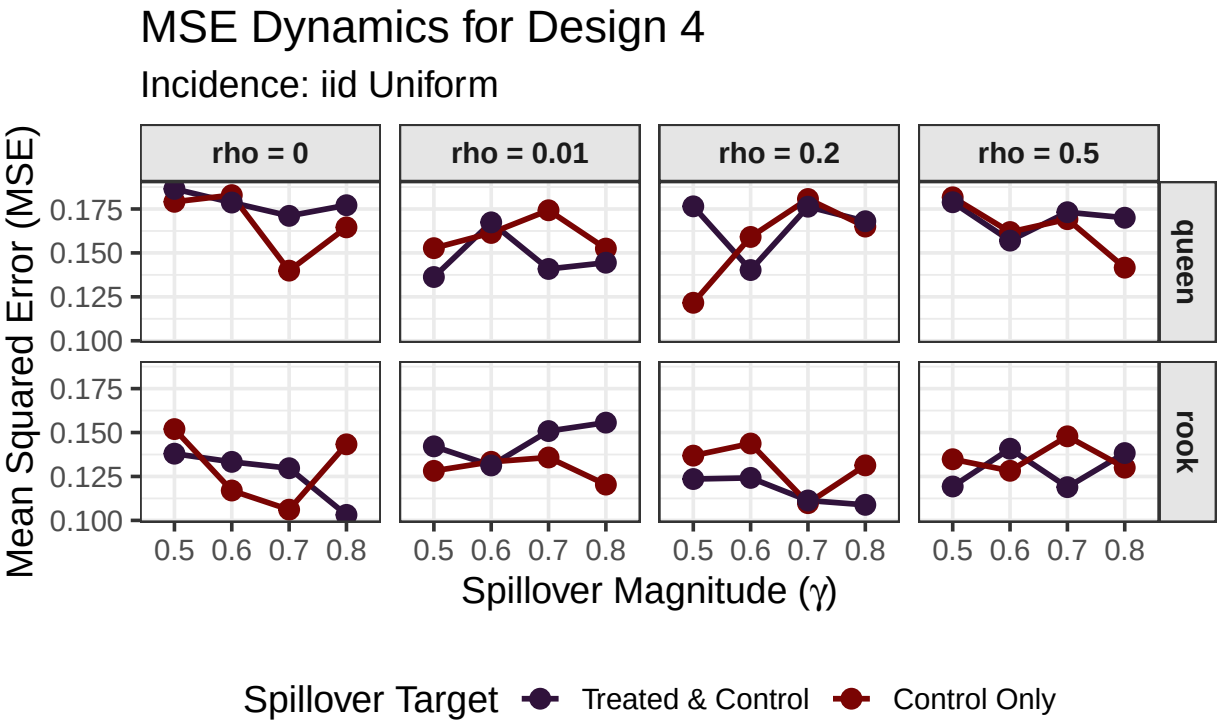

Figure 21: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

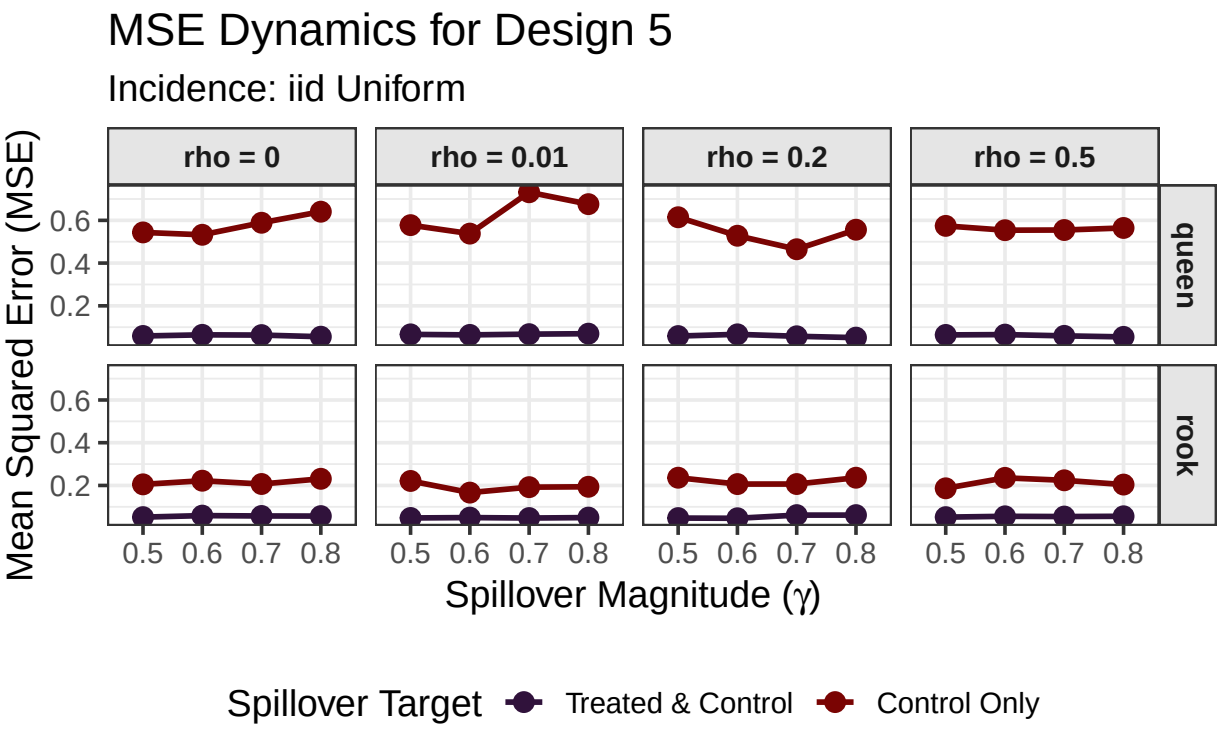

Figure 22: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

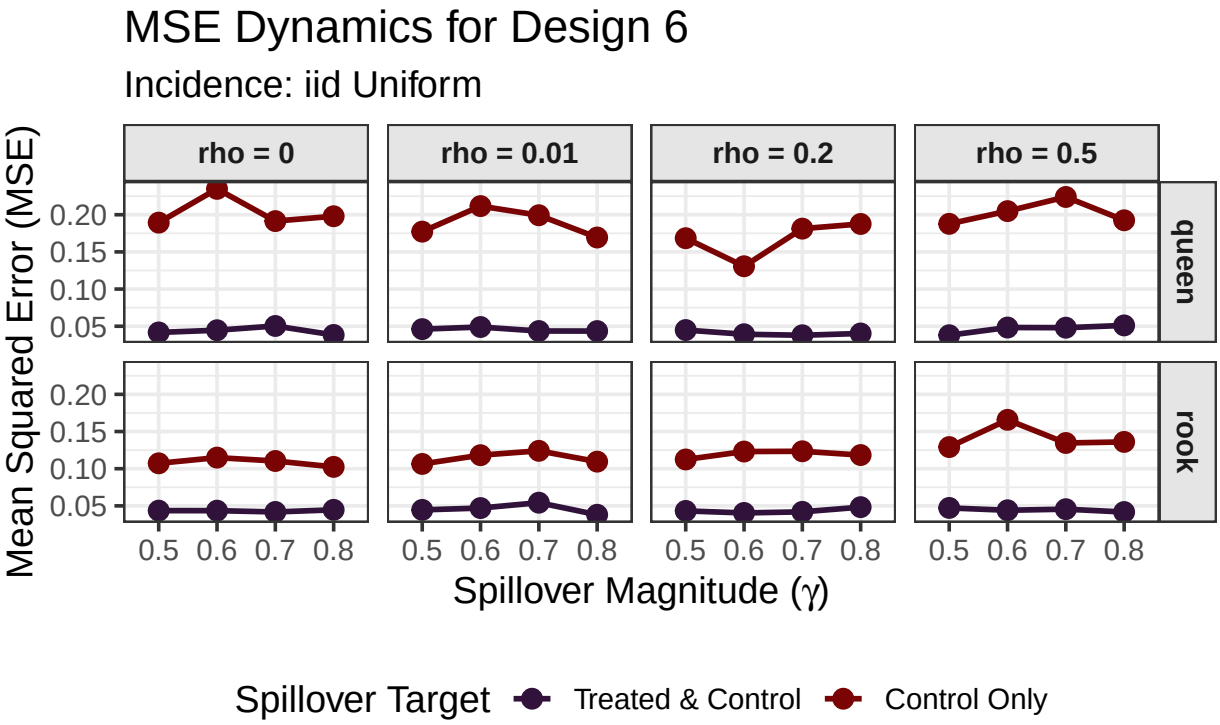

Figure 23: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

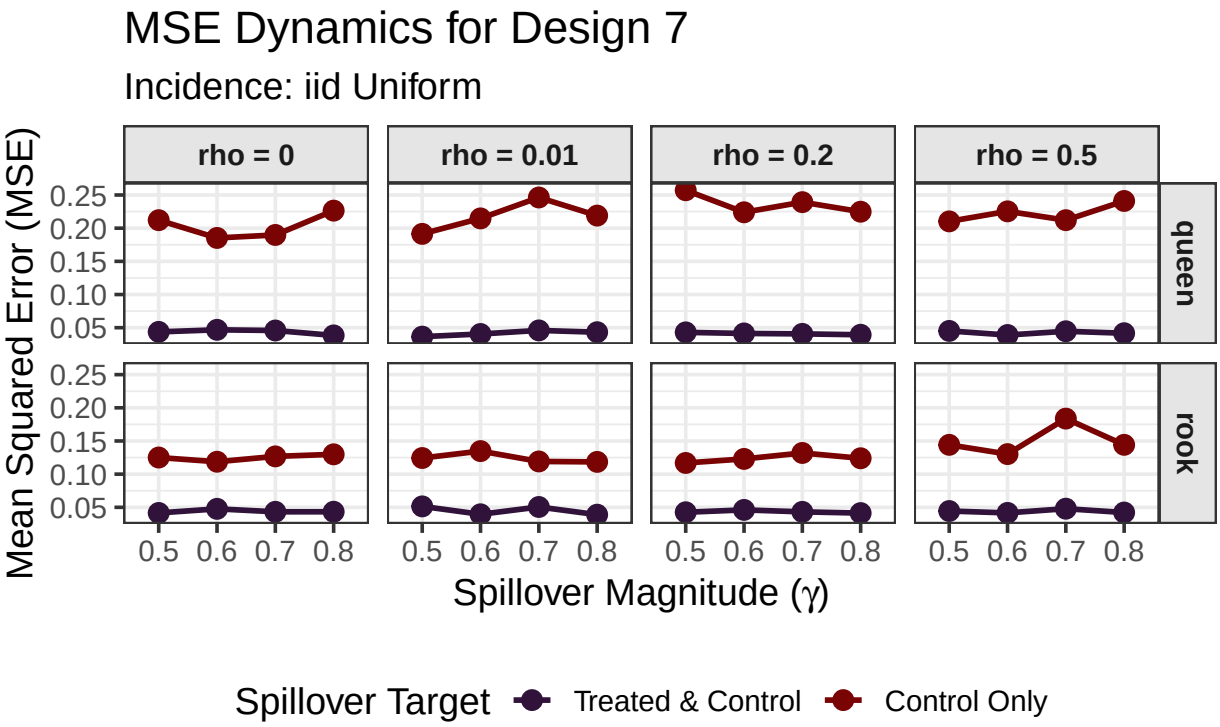

Figure 24: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

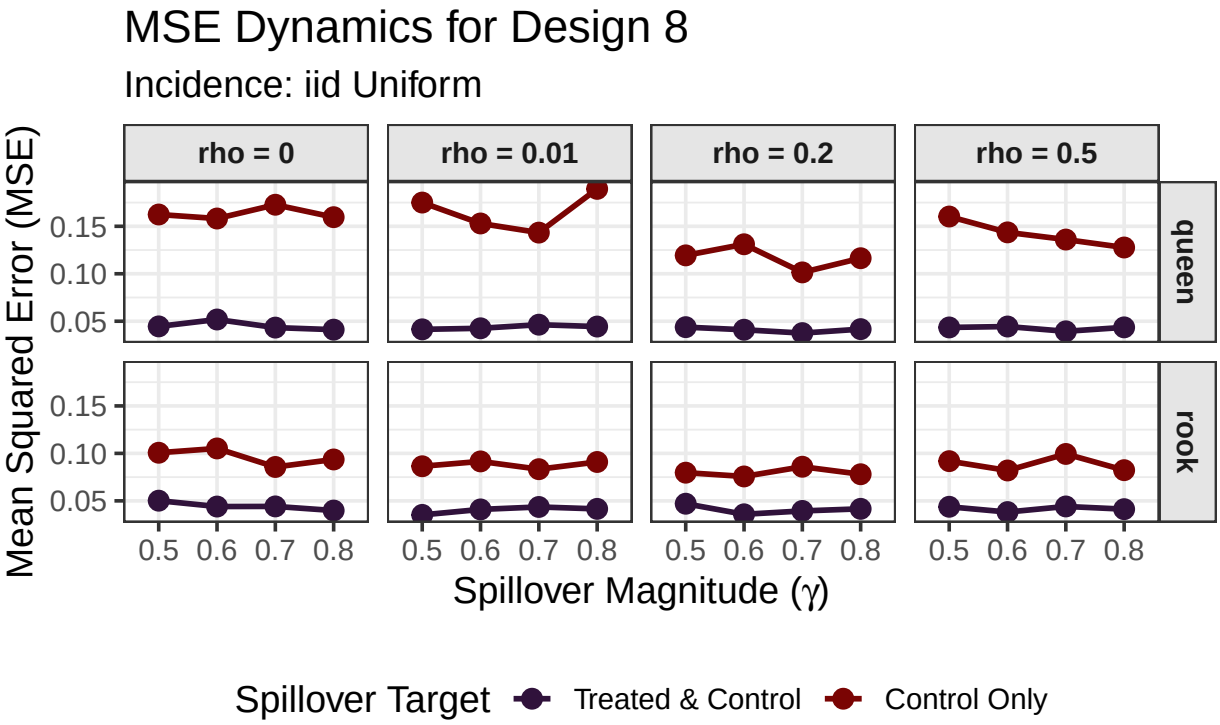

Figure 25: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

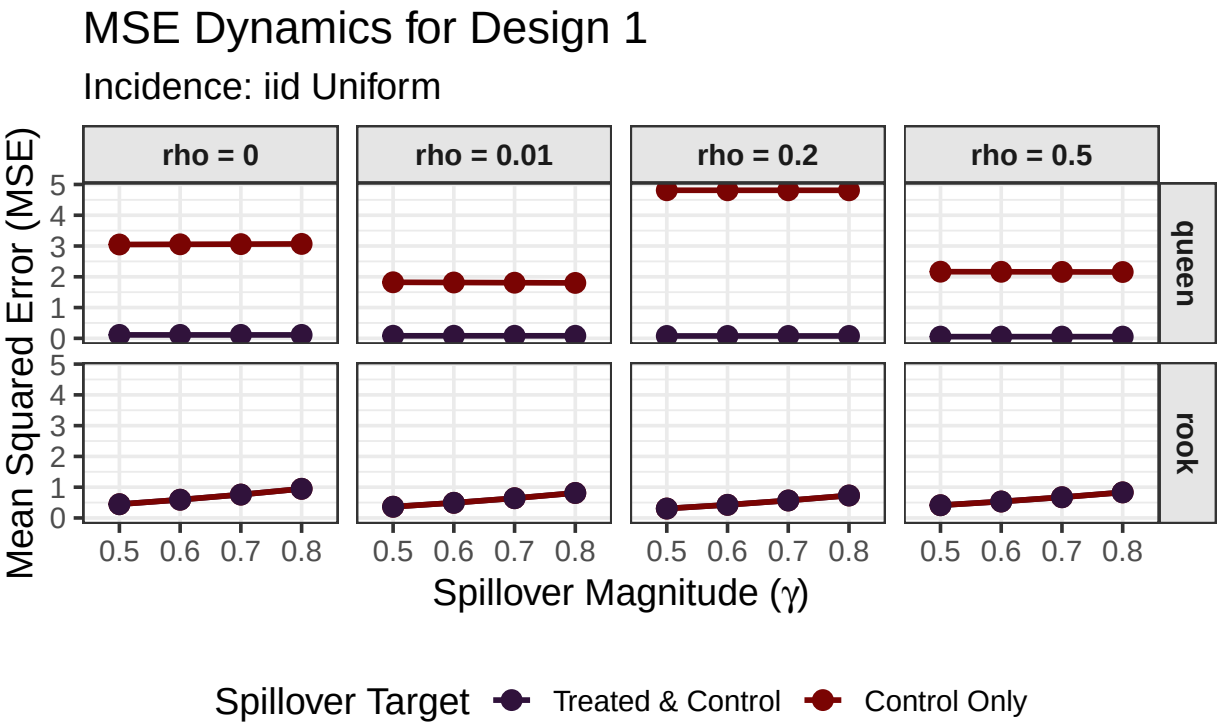

Figure 26: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

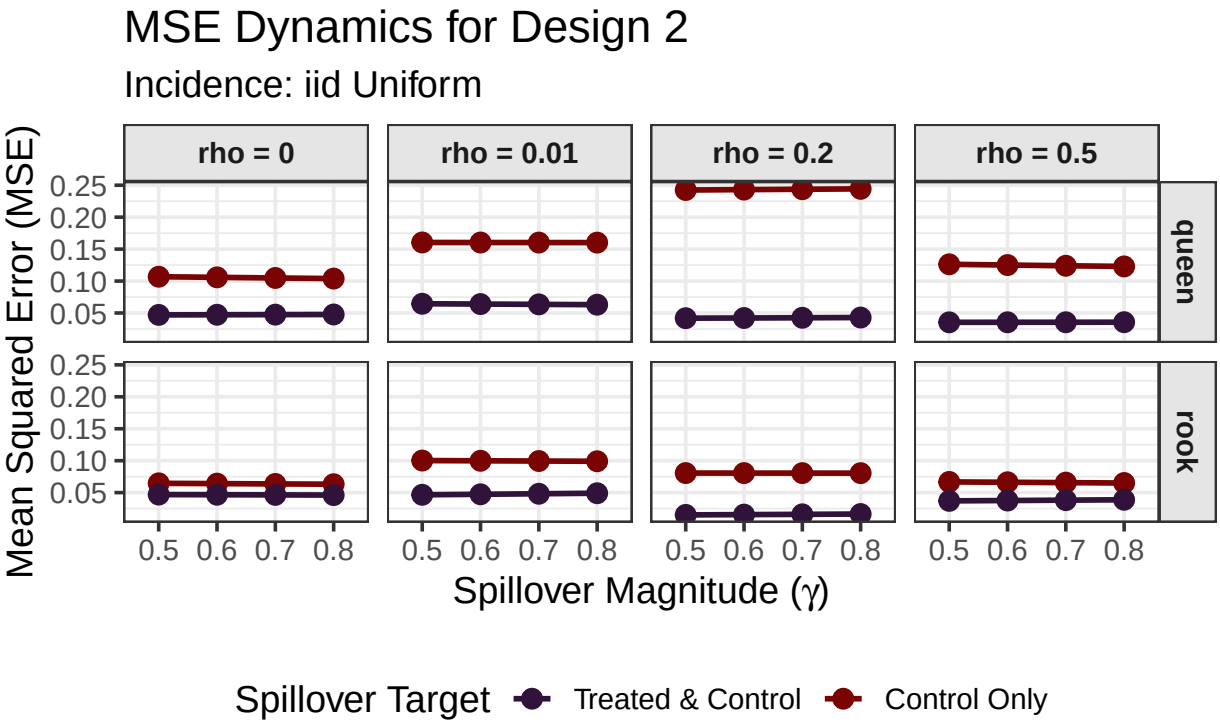

Figure 27: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

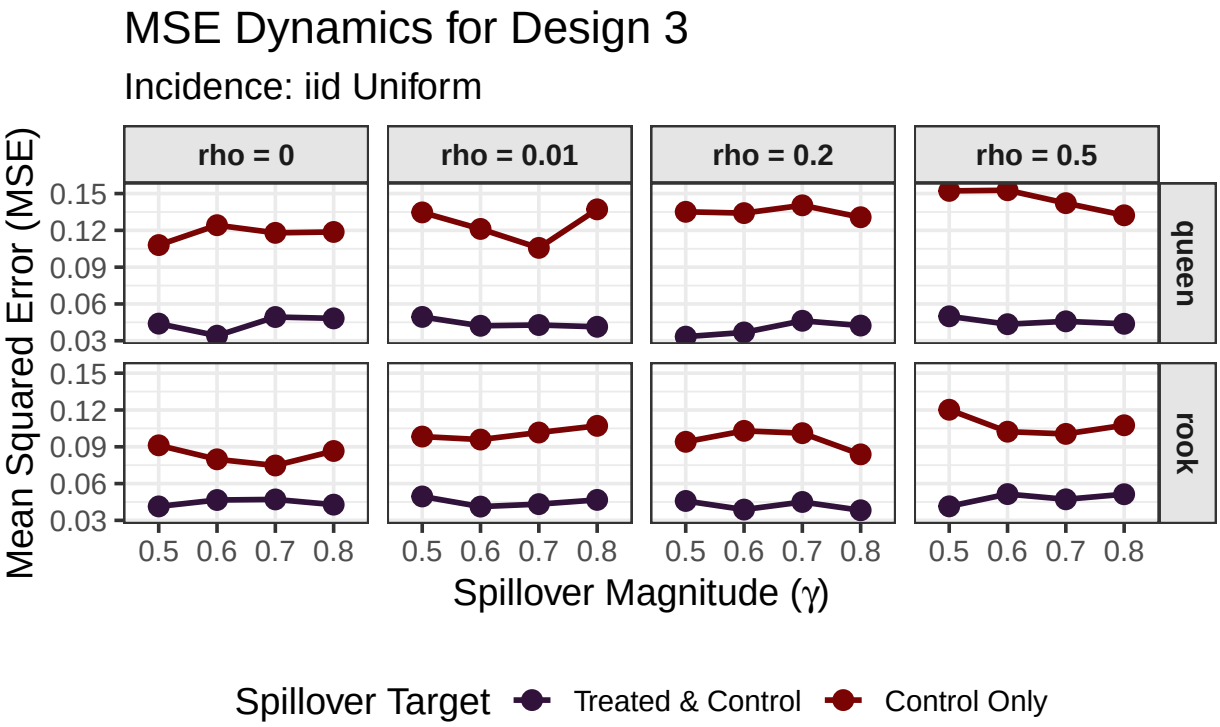

Figure 28: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

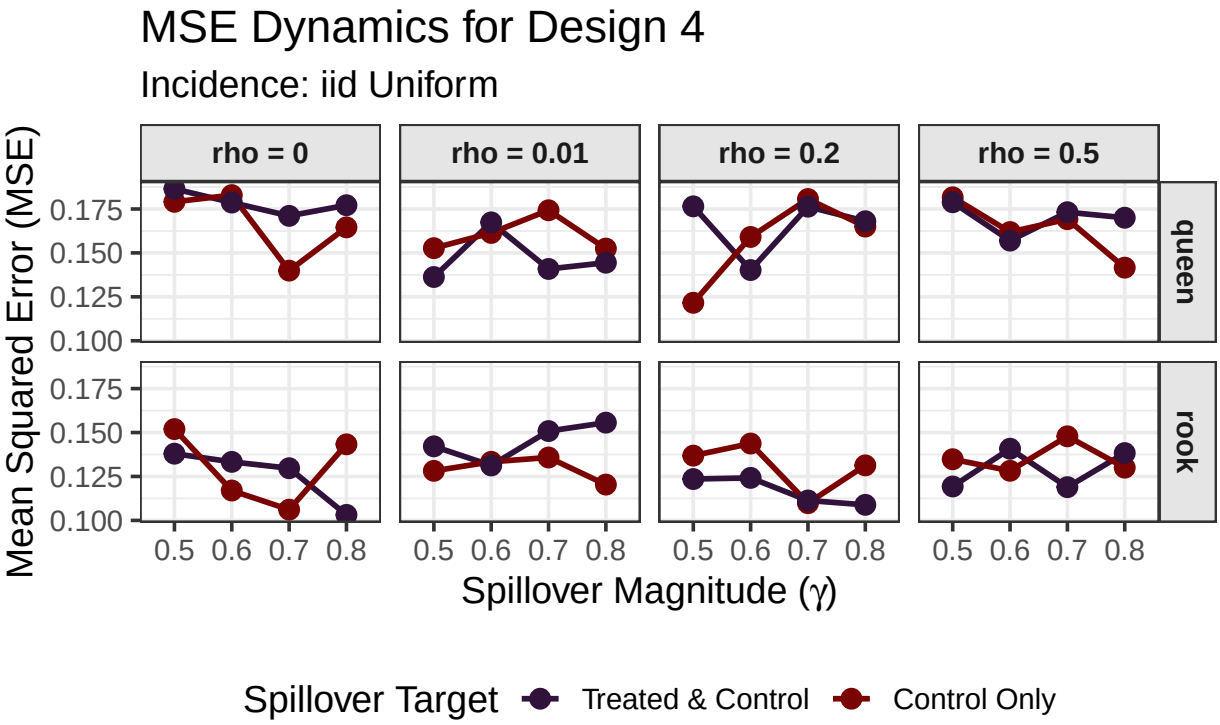

Figure 29: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

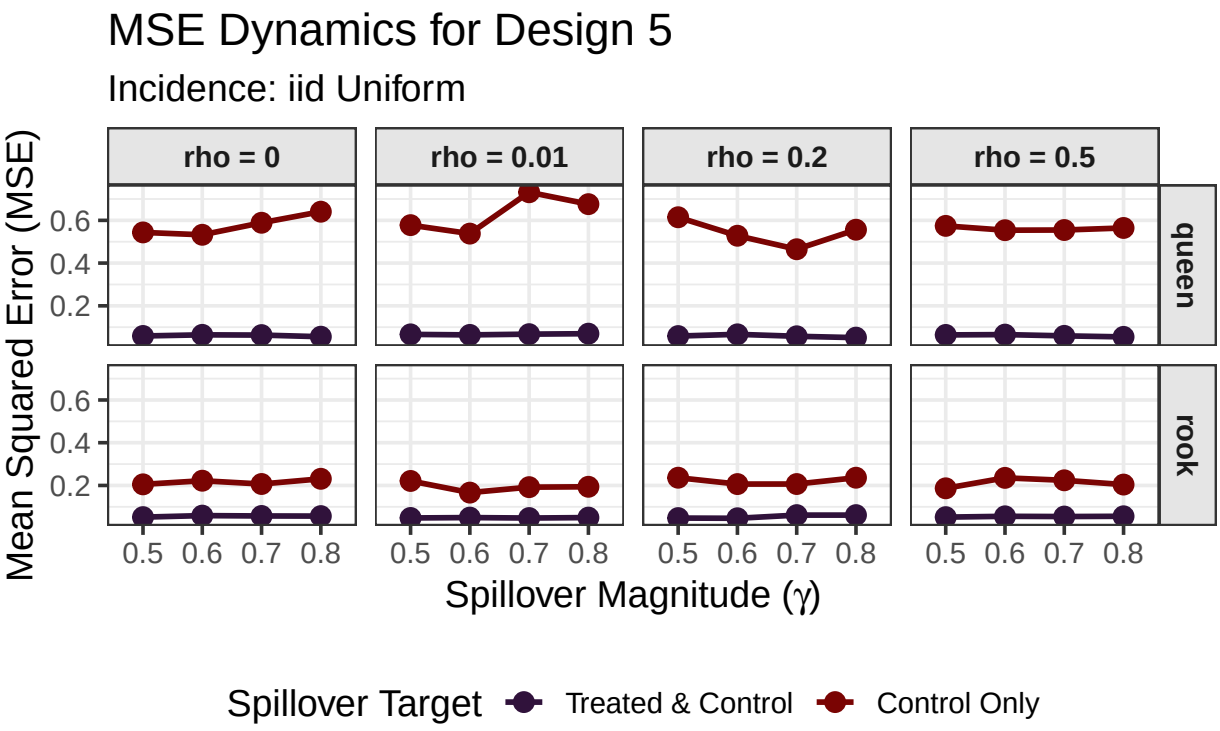

Figure 30: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

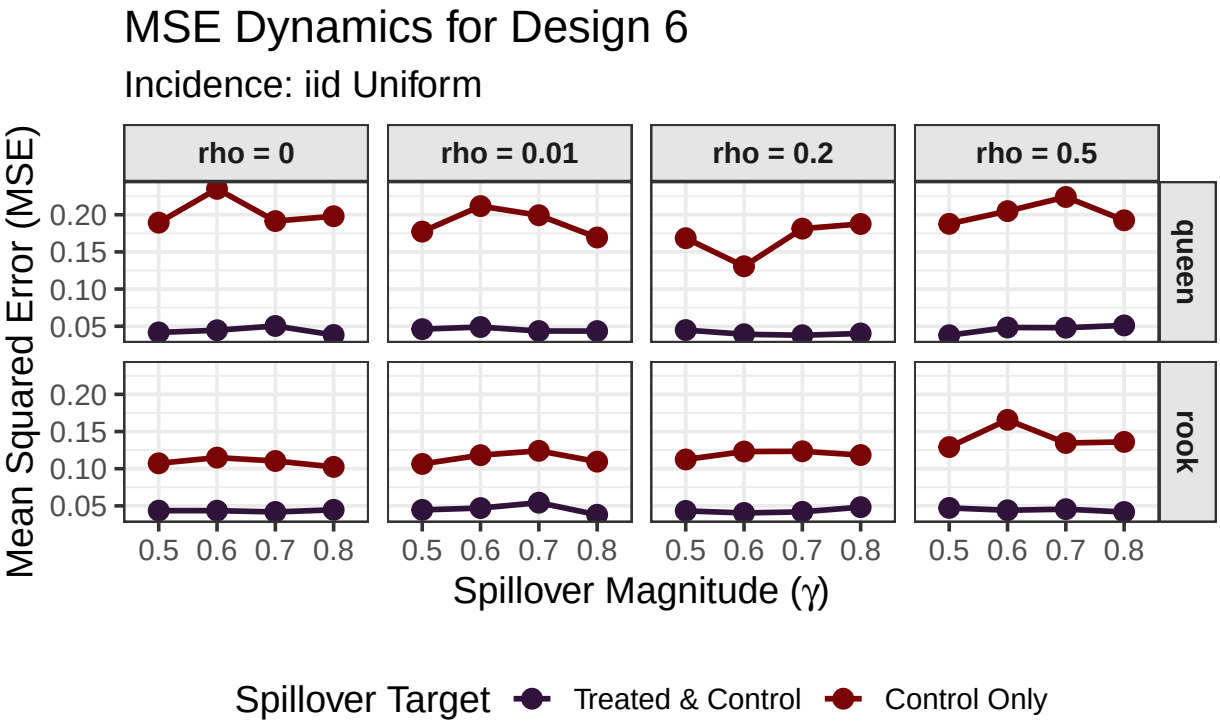

Figure 31: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

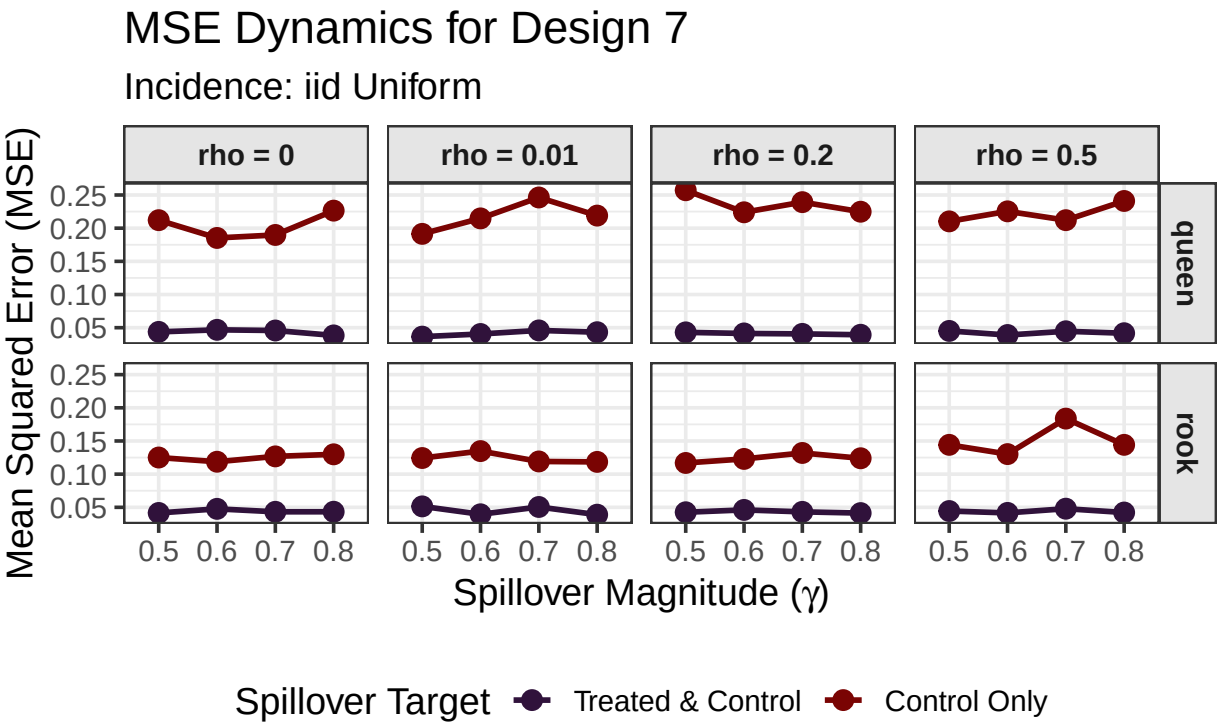

Figure 32: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

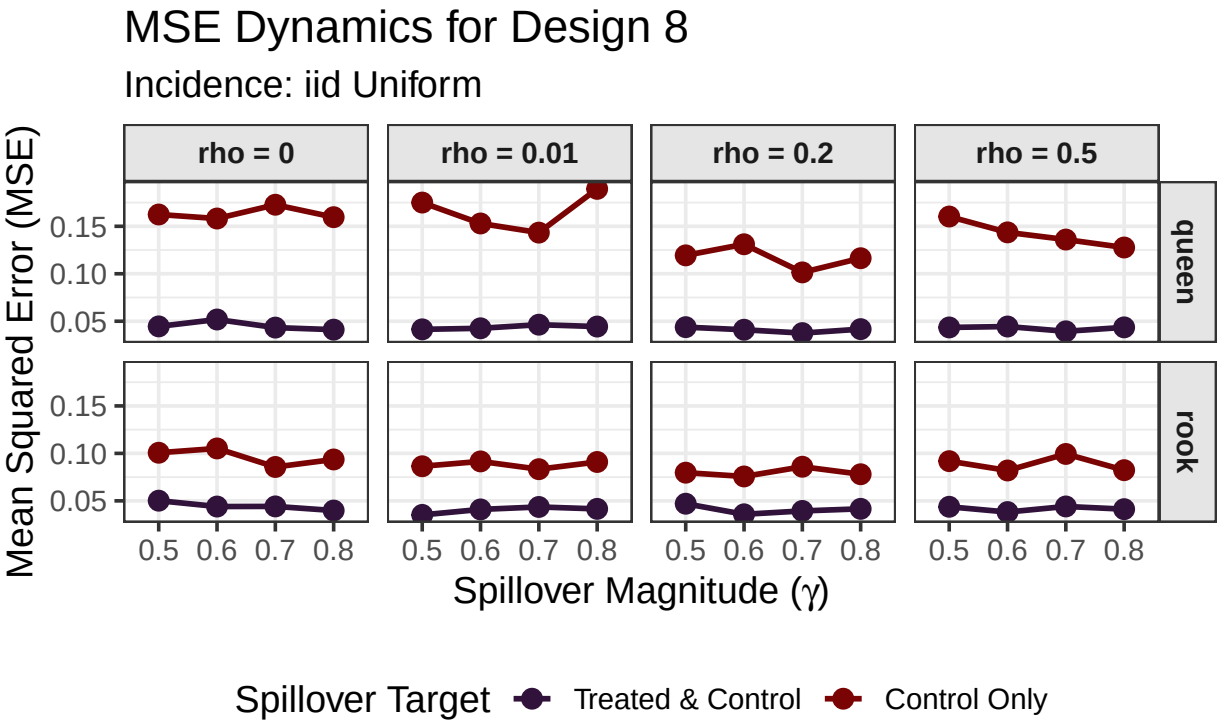

Figure 33: Per-design MSE line plots across gamma levels, for iid Uniform incidence.

12.5 Marginal Ranking Tables

12.5.1 iid Uniform

Table 18: iid Uniform | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	**0.0722 (#2)**	**0.0786 (#1)**	**0.0781 (#2)**	**0.0865 (#3)**
D2: High Inc. Focus	0.0658 (#1)	0.0929 (#3)	0.0955 (#4)	0.0660 (#1)
D8: Inc.-Guided Sat. Q.	0.0873 (#3)	0.0843 (#2)	0.0697 (#1)	0.0788 (#2)
D6: Balanced Quartiles	0.0998 (#4)	0.0989 (#4)	0.0926 (#3)	0.1086 (#4)
D7: Balanced Halves	0.1040 (#5)	0.1072 (#5)	0.1111 (#5)	0.1148 (#5)
D4: Isolation Buffer	0.1501 (#6)	0.1455 (#6)	0.1423 (#6)	0.1495 (#6)
D5: 2x2 Blocking	0.2273 (#7)	0.2349 (#7)	0.2187 (#7)	0.2224 (#7)
D1: Checkerboard	1.1359 (#8)	0.7623 (#8)	1.4759 (#8)	0.8604 (#8)

Table 19: iid Uniform | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	**0.0805 (#2)**	**0.0780 (#1)**	**0.0782 (#2)**	**0.0787 (#1)**
D2: High Inc. Focus	0.0802 (#1)	0.0801 (#3)	0.0800 (#3)	0.0799 (#3)
D8: Inc.-Guided Sat. Q.	0.0828 (#3)	0.0799 (#2)	0.0778 (#1)	0.0796 (#2)
D6: Balanced Quartiles	0.0954 (#4)	0.1037 (#4)	0.1032 (#4)	0.0975 (#4)
D7: Balanced Halves	0.1081 (#5)	0.1062 (#5)	0.1132 (#5)	0.1098 (#5)
D4: Isolation Buffer	0.1492 (#6)	0.1475 (#6)	0.1460 (#6)	0.1446 (#6)
D5: 2x2 Blocking	0.2253 (#7)	0.2158 (#7)	0.2273 (#7)	0.2347 (#7)
D1: Checkerboard	0.9523 (#8)	1.0164 (#8)	1.0906 (#8)	1.1751 (#8)

Table 20: iid Uniform | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	**0.0441 (#4)**	**0.1136 (#1)**
D2: High Inc. Focus	0.0421 (#1)	0.1179 (#3)
D8: Inc.-Guided Sat. Q.	0.0425 (#2)	0.1176 (#2)
D6: Balanced Quartiles	0.0442 (#5)	0.1557 (#5)
D7: Balanced Halves	0.0432 (#3)	0.1754 (#6)
D4: Isolation Buffer	0.1472 (#7)	0.1465 (#4)
D5: 2x2 Blocking	0.0574 (#6)	0.3942 (#7)
D1: Checkerboard	0.3390 (#8)	1.7782 (#8)

12.5.2 Poisson (rho_X = 0.20)

Table 21: Poisson (rho X = 0.20) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	**0.0704 (#1)**	**0.0775 (#1)**	**0.0743 (#1)**	**0.0889 (#1)**
D3: Sat. Quadrants	0.0737 (#2)	0.0835 (#2)	0.0794 (#2)	0.0996 (#2)
D6: Balanced Quartiles	0.1033 (#3)	0.0988 (#3)	0.1044 (#3)	0.1238 (#3)
D7: Balanced Halves	0.1097 (#4)	0.1023 (#4)	0.1085 (#4)	0.1252 (#4)
D4: Isolation Buffer	0.1382 (#5)	0.1373 (#5)	0.1552 (#6)	0.1581 (#6)

D2: High Inc. Focus	0.2770 (#7)	0.2468 (#7)	0.1347 (#5)	0.1484 (#5)
D5: 2x2 Blocking	0.2303 (#6)	0.2183 (#6)	0.2177 (#7)	0.2597 (#7)
D1: Checkerboard	0.9923 (#8)	1.0645 (#8)	1.0058 (#8)	0.6218 (#8)

Table 22: Poisson (rho X = 0.20) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	**0.0769 (#1)**	**0.0792 (#1)**	**0.0756 (#1)**	**0.0793 (#1)**
D3: Sat. Quadrants	0.0847 (#2)	0.0863 (#2)	0.0825 (#2)	0.0827 (#2)
D6: Balanced Quartiles	0.1079 (#3)	0.1031 (#3)	0.1136 (#3)	0.1058 (#3)
D7: Balanced Halves	0.1117 (#4)	0.1069 (#4)	0.1161 (#4)	0.1110 (#4)
D4: Isolation Buffer	0.1394 (#5)	0.1500 (#5)	0.1525 (#5)	0.1468 (#5)
D2: High Inc. Focus	0.2019 (#6)	0.2018 (#6)	0.2017 (#6)	0.2016 (#6)
D5: 2x2 Blocking	0.2408 (#7)	0.2286 (#7)	0.2251 (#7)	0.2315 (#7)
D1: Checkerboard	0.8311 (#8)	0.8845 (#8)	0.9478 (#8)	1.0210 (#8)

Table 23: Poisson (rho X = 0.20) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	**0.0451 (#3)**	**0.1105 (#1)**
D3: Sat. Quadrants	0.0467 (#4)	0.1215 (#2)
D6: Balanced Quartiles	0.0439 (#2)	0.1713 (#4)
D7: Balanced Halves	0.0437 (#1)	0.1791 (#5)
D4: Isolation Buffer	0.1455 (#6)	0.1489 (#3)
D2: High Inc. Focus	0.1895 (#7)	0.2139 (#6)
D5: 2x2 Blocking	0.0594 (#5)	0.4036 (#7)
D1: Checkerboard	0.2712 (#8)	1.5710 (#8)

12.5.3 Poisson (rho_X = 0.50)

Table 24: Poisson (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	**0.0794 (#2)**	**0.0711 (#1)**	**0.0839 (#2)**	**0.0816 (#2)**
D8: Inc.-Guided Sat. Q.	0.0834 (#3)	0.0837 (#2)	0.0792 (#1)	0.0780 (#1)
D6: Balanced Quartiles	0.1000 (#4)	0.0994 (#4)	0.1037 (#3)	0.1067 (#3)
D7: Balanced Halves	0.1100 (#5)	0.0911 (#3)	0.1087 (#4)	0.1080 (#4)
D4: Isolation Buffer	0.1508 (#6)	0.1324 (#6)	0.1462 (#5)	0.1494 (#5)
D2: High Inc. Focus	0.0589 (#1)	0.1264 (#5)	0.2972 (#7)	0.3462 (#7)
D5: 2x2 Blocking	0.2436 (#7)	0.2074 (#7)	0.2448 (#6)	0.2240 (#6)
D1: Checkerboard	0.5263 (#8)	0.7522 (#8)	0.6228 (#8)	0.8764 (#8)

Table 25: Poisson (rho X = 0.50) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	**0.0795 (#2)**	**0.0774 (#1)**	**0.0779 (#1)**	**0.0813 (#2)**
D8: Inc.-Guided Sat. Q.	0.0790 (#1)	0.0813 (#2)	0.0831 (#2)	0.0809 (#1)
D6: Balanced Quartiles	0.1066 (#4)	0.1016 (#3)	0.1017 (#3)	0.0999 (#3)
D7: Balanced Halves	0.1052 (#3)	0.1059 (#4)	0.1029 (#4)	0.1038 (#4)
D4: Isolation Buffer	0.1420 (#5)	0.1461 (#5)	0.1428 (#5)	0.1479 (#5)

D2: High Inc. Focus	0.2065 (#6)	0.2069 (#6)	0.2074 (#6)	0.2079 (#6)
D5: 2x2 Blocking	0.2331 (#7)	0.2308 (#7)	0.2244 (#7)	0.2315 (#7)
D1: Checkerboard	0.6067 (#8)	0.6587 (#8)	0.7204 (#8)	0.7919 (#8)

Table 26: Poisson (rho X = 0.50) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	**0.0460 (#4)**	**0.1121 (#1)**
D8: Inc.-Guided Sat. Q.	0.0449 (#3)	0.1172 (#2)
D6: Balanced Quartiles	0.0430 (#2)	0.1619 (#4)
D7: Balanced Halves	0.0424 (#1)	0.1665 (#5)
D4: Isolation Buffer	0.1453 (#6)	0.1441 (#3)
D2: High Inc. Focus	0.1971 (#7)	0.2173 (#6)
D5: 2x2 Blocking	0.0551 (#5)	0.4048 (#7)
D1: Checkerboard	0.2604 (#8)	1.1285 (#8)

12.5.4 Spatial (rho_X = 0.20)

Table 27: Spatial (rho X = 0.20) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D8: Inc.-Guided Sat. Q.	**0.0863 (#3)**	**0.0667 (#2)**	**0.0796 (#2)**	**0.0786 (#1)**
D3: Sat. Quadrants	0.0771 (#2)	0.0824 (#3)	0.0785 (#1)	0.0860 (#2)
D2: High Inc. Focus	0.0592 (#1)	0.0614 (#1)	0.1303 (#5)	0.1419 (#5)
D6: Balanced Quartiles	0.1012 (#4)	0.0997 (#4)	0.1012 (#3)	0.1005 (#3)
D7: Balanced Halves	0.1055 (#5)	0.1189 (#5)	0.1151 (#4)	0.1121 (#4)
D4: Isolation Buffer	0.1460 (#6)	0.1524 (#6)	0.1537 (#6)	0.1493 (#6)
D5: 2x2 Blocking	0.2272 (#7)	0.2231 (#7)	0.2343 (#7)	0.2317 (#7)
D1: Checkerboard	0.4783 (#8)	0.6645 (#8)	1.1911 (#8)	0.8306 (#8)

Table 28: Spatial (rho X = 0.20) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D8: Inc.-Guided Sat. Q.	**0.0803 (#1)**	**0.0780 (#1)**	**0.0764 (#1)**	**0.0764 (#1)**
D3: Sat. Quadrants	0.0819 (#2)	0.0811 (#2)	0.0818 (#2)	0.0792 (#2)
D2: High Inc. Focus	0.0982 (#3)	0.0982 (#3)	0.0982 (#3)	0.0982 (#3)
D6: Balanced Quartiles	0.0990 (#4)	0.1041 (#4)	0.1011 (#4)	0.0985 (#4)
D7: Balanced Halves	0.1131 (#5)	0.1080 (#5)	0.1166 (#5)	0.1137 (#5)
D4: Isolation Buffer	0.1412 (#6)	0.1488 (#6)	0.1544 (#6)	0.1570 (#6)
D5: 2x2 Blocking	0.2364 (#7)	0.2351 (#7)	0.2135 (#7)	0.2313 (#7)
D1: Checkerboard	0.7063 (#8)	0.7562 (#8)	0.8161 (#8)	0.8860 (#8)

Table 29: Spatial (rho X = 0.20) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D8: Inc.-Guided Sat. Q.	**0.0445 (#3)**	**0.1111 (#1)**
D3: Sat. Quadrants	0.0459 (#4)	0.1161 (#2)
D2: High Inc. Focus	0.0487 (#5)	0.1477 (#3)
D6: Balanced Quartiles	0.0434 (#2)	0.1579 (#5)
D7: Balanced Halves	0.0432 (#1)	0.1826 (#6)

D4: Isolation Buffer	0.1520 (#7)	0.1488 (#4)
D5: 2x2 Blocking	0.0596 (#6)	0.3985 (#7)
D1: Checkerboard	0.2620 (#8)	1.3203 (#8)

12.5.5 Spatial (rho_X = 0.50)

Table 30: Spatial (rho X = 0.50) | Marginal rankings by Rho (outcome spatial autocorrelation).

Design	Rho=0	Rho=0.01	Rho=0.2	Rho=0.5
D3: Sat. Quadrants	**0.0806 (#1)**	**0.0730 (#3)**	**0.0780 (#2)**	**0.0699 (#1)**
D8: Inc.-Guided Sat. Q.	0.0838 (#2)	0.0722 (#2)	0.0748 (#1)	0.0790 (#2)
D2: High Inc. Focus	0.0874 (#3)	0.0623 (#1)	0.0803 (#3)	0.0824 (#3)
D6: Balanced Quartiles	0.0973 (#4)	0.1077 (#4)	0.1052 (#4)	0.0948 (#4)
D7: Balanced Halves	0.1106 (#5)	0.1121 (#5)	0.1100 (#5)	0.1057 (#5)
D4: Isolation Buffer	0.1551 (#6)	0.1482 (#6)	0.1533 (#6)	0.1427 (#6)
D5: 2x2 Blocking	0.2247 (#7)	0.2260 (#7)	0.2238 (#7)	0.2140 (#7)
D1: Checkerboard	0.4400 (#8)	0.4855 (#8)	0.6655 (#8)	0.5936 (#8)

Table 31: Spatial (rho X = 0.50) | Marginal rankings by Gamma (spillover magnitude).

Design	Gamma=0.5	Gamma=0.6	Gamma=0.7	Gamma=0.8
D3: Sat. Quadrants	**0.0698 (#1)**	**0.0778 (#2)**	**0.0757 (#1)**	**0.0781 (#3)**
D8: Inc.-Guided Sat. Q.	0.0777 (#2)	0.0773 (#1)	0.0774 (#2)	0.0774 (#1)
D2: High Inc. Focus	0.0783 (#3)	0.0782 (#3)	0.0780 (#3)	0.0779 (#2)
D6: Balanced Quartiles	0.0965 (#4)	0.1028 (#4)	0.1047 (#5)	0.1010 (#4)
D7: Balanced Halves	0.1100 (#5)	0.1107 (#5)	0.1029 (#4)	0.1149 (#5)
D4: Isolation Buffer	0.1533 (#6)	0.1463 (#6)	0.1508 (#6)	0.1488 (#6)
D5: 2x2 Blocking	0.2214 (#7)	0.2332 (#7)	0.2241 (#7)	0.2099 (#7)
D1: Checkerboard	0.4539 (#8)	0.5084 (#8)	0.5734 (#8)	0.6489 (#8)

Table 32: Spatial (rho X = 0.50) | Marginal rankings by Spillover Type.

Design	Spillover_Type=both	Spillover_Type=control_only
D3: Sat. Quadrants	**0.0426 (#2)**	**0.1081 (#1)**
D8: Inc.-Guided Sat. Q.	0.0455 (#5)	0.1094 (#2)
D2: High Inc. Focus	0.0441 (#4)	0.1121 (#3)
D6: Balanced Quartiles	0.0432 (#3)	0.1593 (#5)
D7: Balanced Halves	0.0423 (#1)	0.1769 (#6)
D4: Isolation Buffer	0.1497 (#7)	0.1499 (#4)
D5: 2x2 Blocking	0.0558 (#6)	0.3885 (#7)
D1: Checkerboard	0.2754 (#8)	0.8169 (#8)

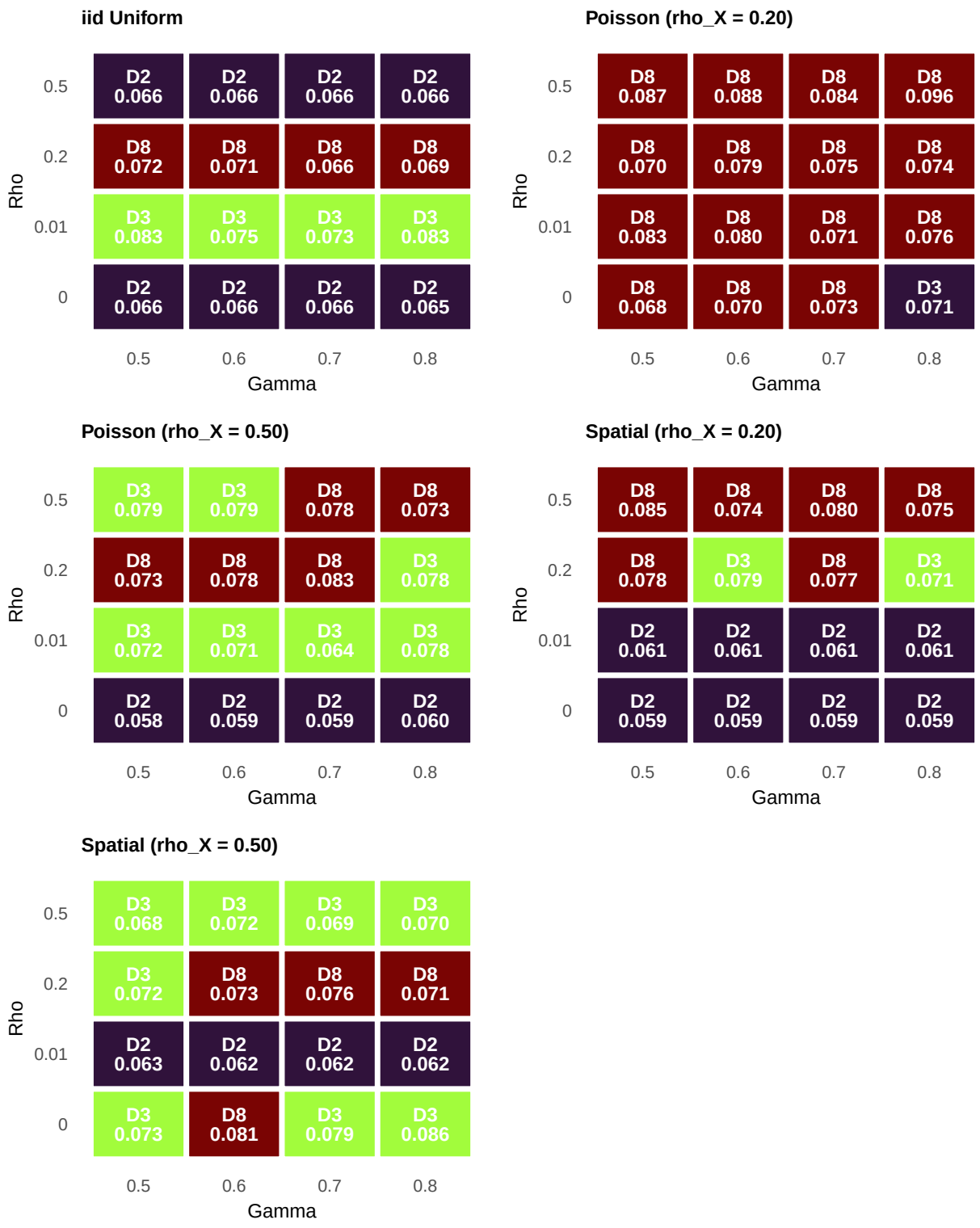

Figure 34: Best design heatmap: Rho vs. Gamma for each incidence config. Each tile shows which design achieves the lowest average MSE.

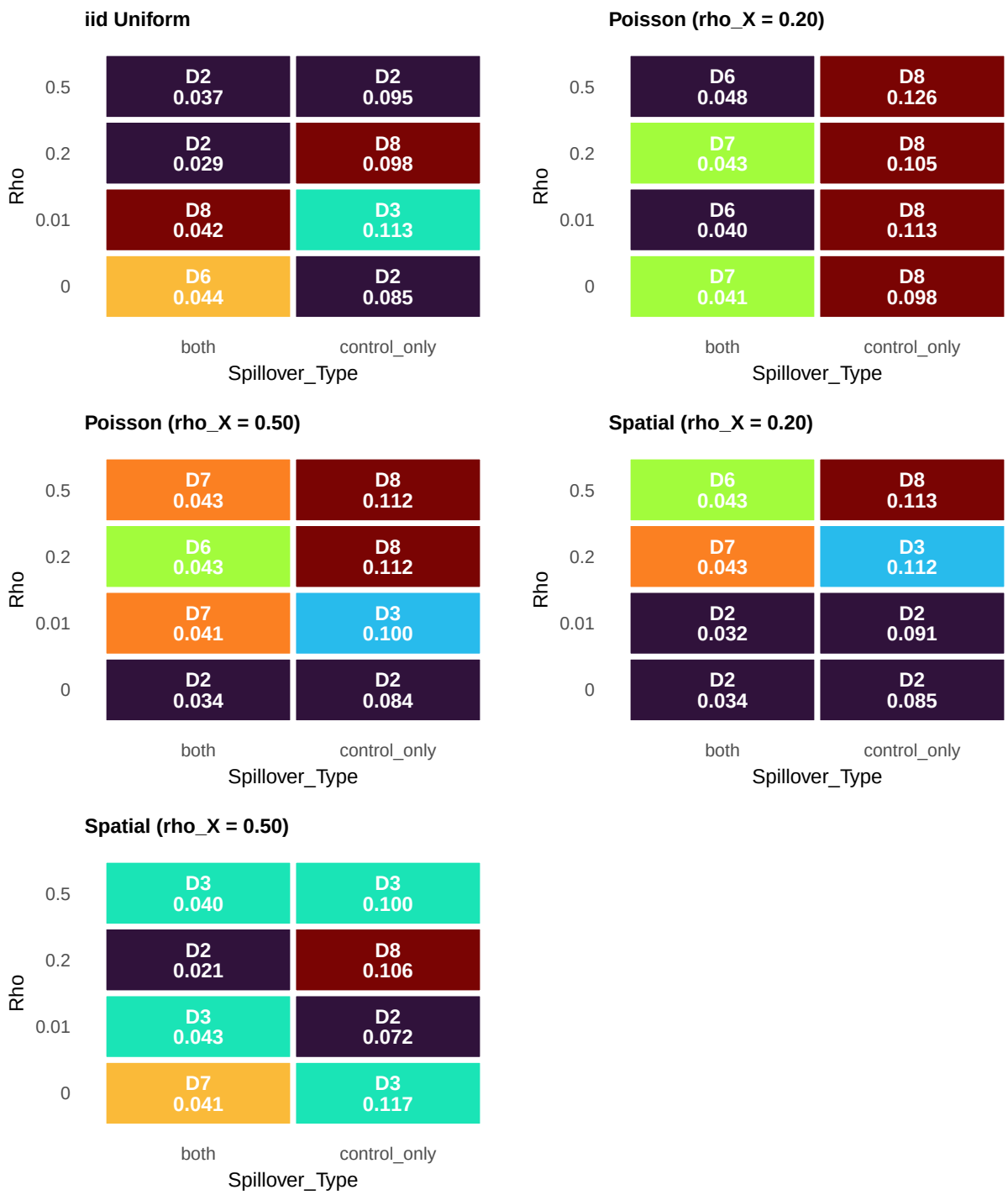

Figure 35: Best design heatmap: Rho vs. Spillover Type for each incidence config. Reveals the key spillover-type interaction.

13 Results: Best Design by Parameter Combination (Q3)

13.1 Rho x Gamma Best-Design Maps

13.2 Rho x Spillover Type Best-Design Maps

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14 Per-Config Narrative Findings

14.1 iid Uniform

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0788). Runner-up: D2: High Inc. Focus (0.0800). The winner dominates in 39.1% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.940; worst design (D1: Checkerboard) achieves only 0.469.

Recommendation: Under iid Uniform incidence, select D3: Sat. Quadrants. It provides 13.4x lower MSE than D1: Checkerboard and near-nominal coverage.

14.2 Poisson (rho_X = 0.20)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0778). Runner-up: D3: Sat. Quadrants (0.0841). The winner dominates in 59.4% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. holds rank 1 at every level of Rho and Gamma.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.939; worst design (D1: Checkerboard) achieves only 0.512.

Recommendation: Under Poisson (rho_X = 0.20) incidence, select D8: Inc.-Guided Sat. Q.. It provides 11.8x lower MSE than D1: Checkerboard and near-nominal coverage.

14.3 Poisson (rho_X = 0.50)

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0790). Runner-up: D8: Inc.-Guided Sat. Q. (0.0811). The winner dominates in 54.7% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.939; worst design (D1: Checkerboard) achieves only 0.556.

Recommendation: Under Poisson (rho_X = 0.50) incidence, select D3: Sat. Quadrants. It provides 8.8x lower MSE than D1: Checkerboard and near-nominal coverage.

14.4 Spatial (rho_X = 0.20)

Overall Winner: D8: Inc.-Guided Sat. Q. achieves the lowest average MSE (0.0778). Runner-up: D3: Sat. Quadrants (0.0810). The winner dominates in 57.8% of individual scenarios.

Robustness: D8: Inc.-Guided Sat. Q. shows some rank variation across parameter levels.

Coverage: D8: Inc.-Guided Sat. Q. achieves coverage of 0.948; worst design (D1: Checkerboard) achieves only 0.522.

Recommendation: Under Spatial (rho_X = 0.20) incidence, select D8: Inc.-Guided Sat. Q.. It provides 10.2x lower MSE than D1: Checkerboard and near-nominal coverage.

14.5 Spatial (rho_X = 0.50)

Overall Winner: D3: Sat. Quadrants achieves the lowest average MSE (0.0754). Runner-up: D8: Inc.-Guided Sat. Q. (0.0774). The winner dominates in 57.8% of individual scenarios.

Robustness: D3: Sat. Quadrants shows some rank variation across parameter levels.

Coverage: D3: Sat. Quadrants achieves coverage of 0.945; worst design (D1: Checkerboard) achieves only 0.575.

Recommendation: Under Spatial (rho_X = 0.50) incidence, select D3: Sat. Quadrants. It provides 7.2x lower MSE than D1: Checkerboard and near-nominal coverage.

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15 Practical Design Selection Guide

Table 33: Practical design selection guide.

Situation	Recommended Design	Justification
Default (uncertain parameters)	**D3: Sat. Quadrants**	**Ranks 1st in 3 of 5 configs across all Gamma levels**
Spillover is 'both' AND Rho <= 0.01	D7: Balanced Halves	Winner shifts at this corner; verify from Rho x Spillover_Type heatmap
ANY scenario	Avoid D1: Checkerboard	Invalid CIs (coverage 0.47–0.58); MSE 7.2–13.4x higher than winner
ANY scenario	Avoid D5: 2x2 Blocking under queen contiguity	Queen contiguity amplifies 2x2 block effects

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16 Subgroup Robustness Profiles

16.1 iid Uniform

Table 34: MLE robustness profile: iid Uniform

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0333	0.0439	0.0631	0.1077	0.1527	0.0637
Design 8	0.0352	0.0431	0.0638	0.1024	0.1894	0.0593
Design 2	0.0151	0.0453	0.0638	0.1011	0.2443	0.0558
Design 6	0.0378	0.0440	0.0783	0.1434	0.2348	0.0995
Design 7	0.0365	0.0431	0.0841	0.1841	0.2572	0.1410
Design 5	0.0464	0.0573	0.1180	0.2934	0.7316	0.2360
Design 4	0.1031	0.1309	0.1427	0.1675	0.1864	0.0366
Design 1	0.0552	0.2573	0.5803	1.1608	4.8117	0.9034

16.2 Poisson (rho_X = 0.20)

Table 35: MLE robustness profile: Poisson (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0344	0.0447	0.0591	0.1168	0.1635	0.0721
Design 3	0.0320	0.0461	0.0720	0.1279	0.1873	0.0818
Design 6	0.0332	0.0439	0.0764	0.1792	0.2743	0.1353
Design 7	0.0323	0.0432	0.0789	0.1763	0.2953	0.1331
Design 5	0.0395	0.0583	0.1265	0.2999	0.8210	0.2416
Design 2	0.0503	0.0745	0.1298	0.2812	0.5039	0.2068
Design 4	0.1028	0.1343	0.1474	0.1595	0.1923	0.0253
Design 1	0.0385	0.1816	0.4349	0.9152	3.2188	0.7336

16.3 Poisson (rho_X = 0.50)

Table 36: MLE robustness profile: Poisson (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0365	0.0443	0.0654	0.1191	0.1560	0.0747
Design 3	0.0391	0.0458	0.0677	0.1044	0.1585	0.0586
Design 7	0.0347	0.0426	0.0780	0.1564	0.2555	0.1138
Design 6	0.0378	0.0429	0.0781	0.1524	0.2452	0.1094
Design 5	0.0413	0.0561	0.1209	0.3093	0.7192	0.2533
Design 4	0.1094	0.1287	0.1422	0.1544	0.2044	0.0257
Design 2	0.0255	0.0914	0.1715	0.3540	0.4708	0.2626
Design 1	0.0331	0.1917	0.4525	0.8916	2.5868	0.6999

16.4 Spatial (rho_X = 0.20)

Table 37: MLE robustness profile: Spatial (rho X = 0.20)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 3	0.0377	0.0455	0.0597	0.1047	0.1759	0.0592
Design 8	0.0369	0.0440	0.0674	0.0952	0.1722	0.0512
Design 6	0.0369	0.0429	0.0736	0.1538	0.2587	0.1109
Design 2	0.0269	0.0362	0.0829	0.1186	0.2510	0.0825
Design 7	0.0375	0.0421	0.0832	0.1671	0.2928	0.1250
Design 5	0.0407	0.0597	0.1280	0.3140	0.6820	0.2543
Design 4	0.1018	0.1300	0.1482	0.1731	0.2057	0.0431
Design 1	0.0367	0.2021	0.4248	0.7776	3.6415	0.5756

16.5 Spatial (rho_X = 0.50)

Table 38: MLE robustness profile: Spatial (rho X = 0.50)

Design	Best_MSE	Q25_MSE	Median_MSE	Q75_MSE	Worst_MSE	IQR_MSE
Design 8	0.0360	0.0463	0.0581	0.1057	0.1812	0.0595
Design 3	0.0359	0.0426	0.0615	0.1045	0.1519	0.0619
Design 2	0.0125	0.0479	0.0670	0.1076	0.1563	0.0598
Design 6	0.0371	0.0420	0.0780	0.1428	0.2452	0.1007
Design 7	0.0343	0.0417	0.0801	0.1597	0.2806	0.1180
Design 5	0.0401	0.0565	0.1147	0.3213	0.6491	0.2648
Design 4	0.1106	0.1332	0.1484	0.1667	0.1929	0.0336
Design 1	0.0478	0.2311	0.4729	0.7049	1.6899	0.4739

16.6 MSE Heatmap

16.7 Design Rank Frequency

Table 39: MLE: Design rank frequency (%) for iid Uniform incidence. Each cell = % of scenarios where design achieved that rank.

Design	Rank_6	Rank_7	Rank_8	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5
Design 2	6.2	0.0	0	45.3	9.4	12.5	10.9	15.6
Design 8	0.0	0.0	0	21.9	32.8	26.6	12.5	6.2
Design 3	0.0	0.0	0	17.2	26.6	31.2	12.5	12.5
Design 6	7.8	0.0	0	9.4	10.9	10.9	29.7	31.2
Design 7	28.1	0.0	0	6.2	15.6	12.5	10.9	26.6
Design 1	4.7	20.3	75	0.0	0.0	0.0	0.0	0.0
Design 4	12.5	25.0	25	0.0	4.7	4.7	21.9	6.2

Design	Rank_6	Rank_7	Rank_8	Rank_1	Rank_2	Rank_3	Rank_4	Rank_5
Design 5	40.6	54.7	0	0.0	0.0	1.6	1.6	1.6

---

17 Conclusions and Future Work

The simulation study evaluated 8 treatment assignment designs across 2,560 parameter scenarios.

Key findings:

1. **MLE achieves near-nominal coverage.** Average coverage across all scenarios is 0.89, close to the nominal 0.95 target.
2. **Design 8 is the top performer.** It achieves the lowest average MSE (0.079) and coverage of 0.941.
3. **Design 1 is consistently worst.** Highest MSE (0.802) and lowest coverage (0.527). Its perfect spatial alternation creates systematic confounding.
4. **Spillover type is the key moderator.** Under control-only spillover, D3: Sat. Quadrants dominates. Under bidirectional spillover, D7: Balanced Halves becomes competitive.
5. **Design rankings are stable across incidence modes and Gamma levels.** The ordering is largely consistent, and spillover magnitude does not affect design choice.

17.1 Limitations and Planned Extensions

- **Oracle MLE:** The MLE model uses the true spillover covariate.
- **Tau sensitivity:** τ was swept across $\{0.8, 1.0, 1.5, 2.0, 3.0\}$ in the completed tau-sweep run (12,800 scenarios). See Section 18 for power curves and rank-stability results. Primary results use $\tau = 1.0$.
- **Equal populations:** Poisson mode uses $P_i = 1{,}000$ for all clusters.
- **Grid sensitivity:** Results use $G = 10$ ($N = 100$) only.

---

18 Tau Sensitivity Analysis

The completed tau-sweep simulation (12,800 scenarios) swept $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$. This section asks: do the design rankings established at $\tau = 1.0$ hold across other effect sizes, and when do designs achieve adequate statistical power?

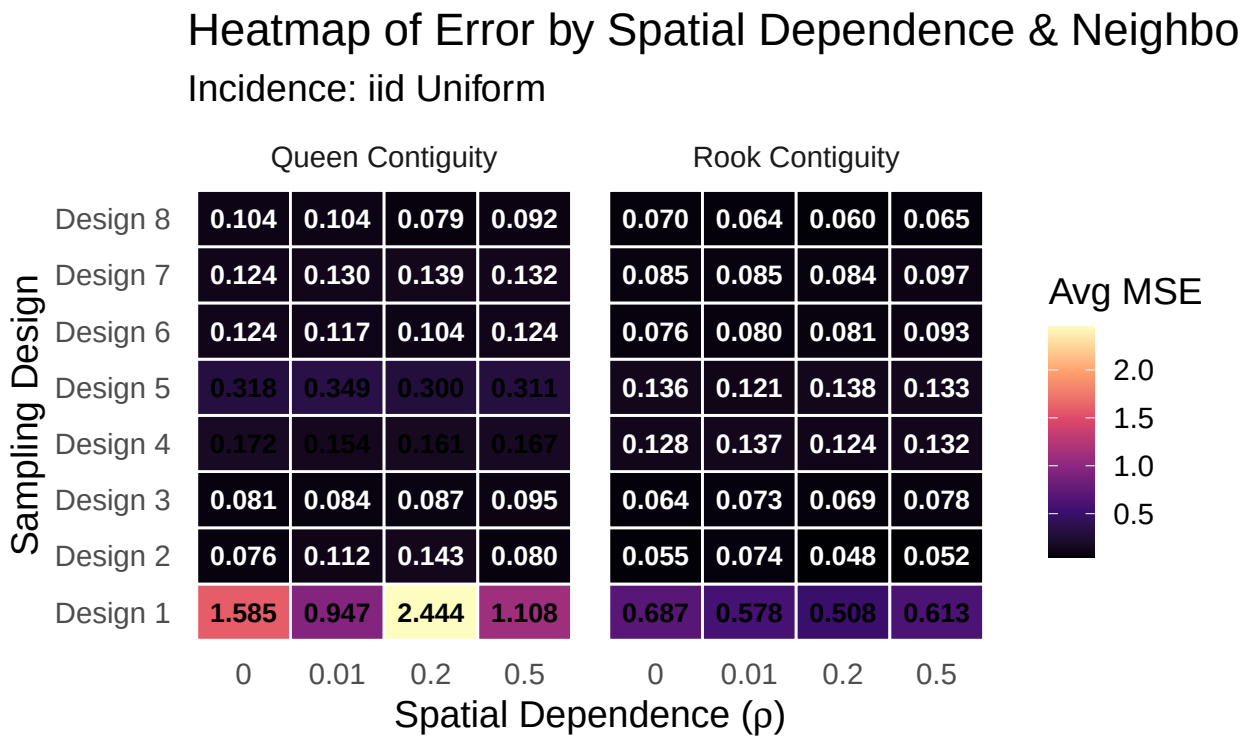

Figure 36: MLE MSE heatmap for iid Uniform incidence.

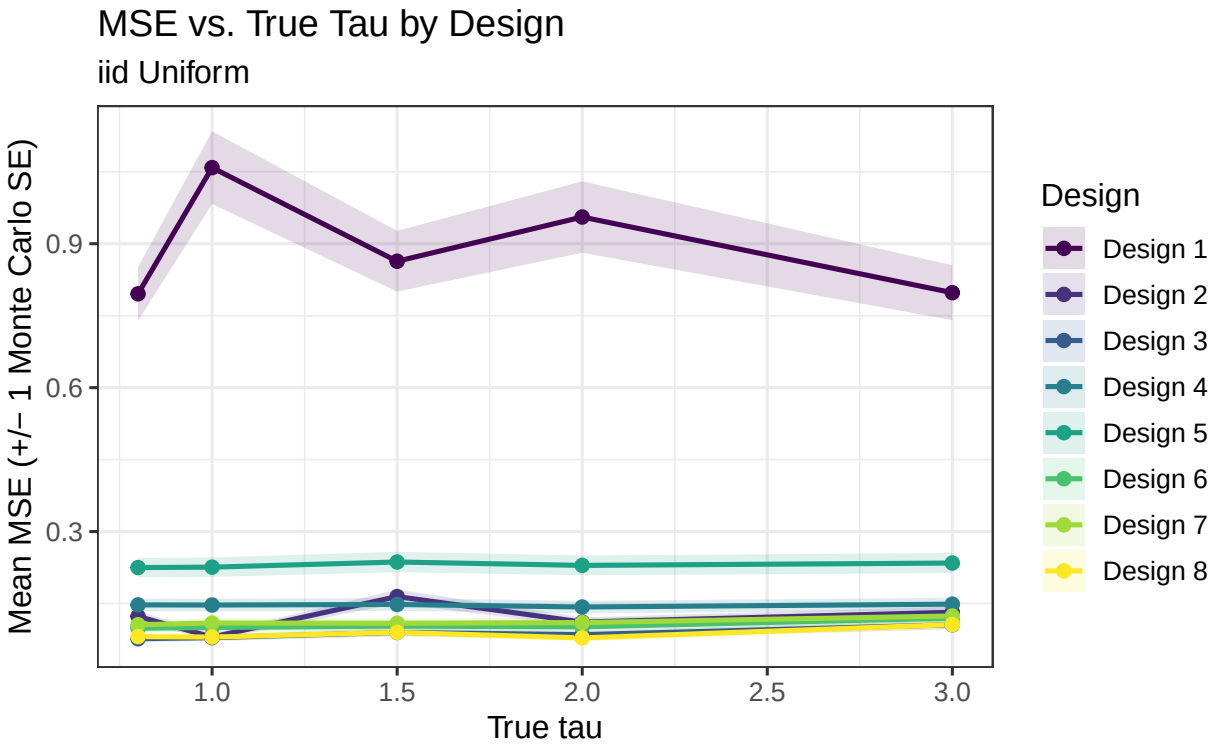

Figure 37: MSE by design across tau values (iid Uniform incidence). Design rankings should be stable; D1 (Checkerboard) is consistently worst.

18.1 MSE Across Tau Values

18.2 Power Curves

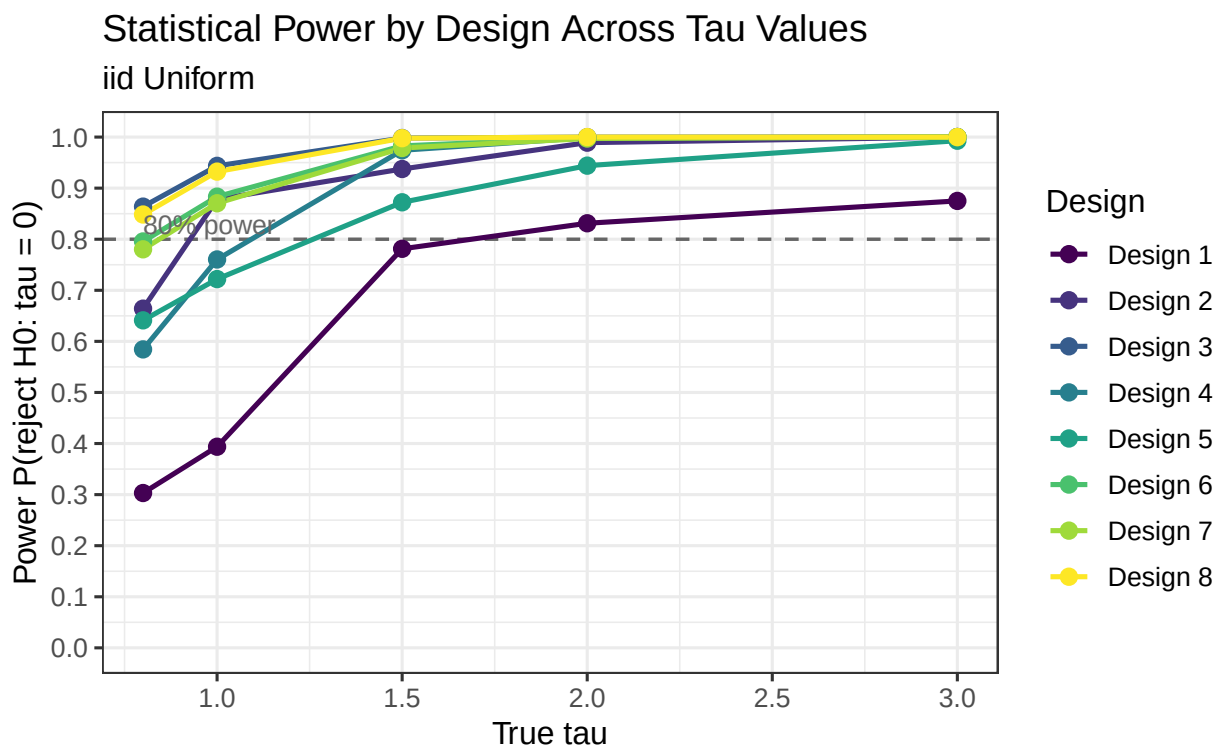

Figure 38: Statistical power P(reject H0: tau=0) by design across tau values (iid Uniform). 80% power reference shown. Efficient designs reach 80% power at smaller effect sizes.

18.3 Coverage Across Tau Values

18.4 Rank Stability Across Tau Values

18.5 Monte Carlo Standard Errors

The `N_Valid_Est` column records the number of non-NA MLE estimates per scenario, serving as the denominator for Monte Carlo (MC) standard errors. All 12,800 scenarios have `N_Valid_Est = 250` (full convergence), confirming that reported MSE and coverage differences reflect genuine design properties rather than MLE instability.

Table 40: Monte Carlo standard errors for MSE at primary scenario (tau=1.0). MC SEs are uniformly small, confirming that MSE differences across designs are not attributable to simulation noise.

Design	Mean N Valid	Mean MSE	MC SE (MSE)
Design 8	250	0.0788	0.00706

Design 3	250	0.0797	0.00715
Design 6	250	0.1024	0.00919
Design 7	250	0.1095	0.00983
Design 2	250	0.1330	0.01181
Design 4	250	0.1478	0.01327
Design 5	250	0.2277	0.02044
Design 1	250	0.8023	0.05760

---

19 Appendix A: Simulation Pseudocode

```
BUILD spatial grid -> coords, W_rook, W_queen, listw_rook, listw_queen
SET I_mat = diag(N)

FOR EACH incidence_mode IN {iid, spatial, poisson}:
  SET rho_X_values = {0} if iid, else {0.20, 0.50}

  FOR EACH rho_X IN rho_X_values:
    GENERATE X_matrix (N x R_o) using incidence_mode, rho_X, W_queen
    SET base_incidence = X_matrix[, 1]

    FOR EACH nb_type IN {rook, queen}:
      SELECT active W, listw for nb_type

      FOR EACH rho IN {0.00, 0.01, 0.20, 0.50}:
        COMPUTE inv_mat = solve(I_mat - rho * W)
        GENERATE Epsilon_matrix (N x R_o) ~ N(0, sigma^2)
        COMPUTE baseline_part = inv_mat %*% (X_matrix * beta + Epsilon_matrix)

        FOR EACH gamma IN {0.50, 0.60, 0.70, 0.80}:
          FOR EACH spill_type IN {control_only, both}:
            FOR EACH design_id IN {1, 2, 3, 4, 5, 6, 7, 8}:

              SET scenario_seed = hash(incidence_mode, rho_X, nb_type,
                                       rho, gamma, spill_type, design_id)
              set.seed(scenario_seed)

              GENERATE Z_matrix (N x R_d)
              FOR d IN 1:R_d:
                Z = Z_matrix[, d]
                COMPUTE spillover = W %*% Z (apply spill_type mask)
                COMPUTE trt_effect = inv_mat %*% (tau * Z + spillover)
                Y_sim = trt_effect + baseline_part
                ESTIMATE tau_hat, CI for each of R_o outcome columns
```

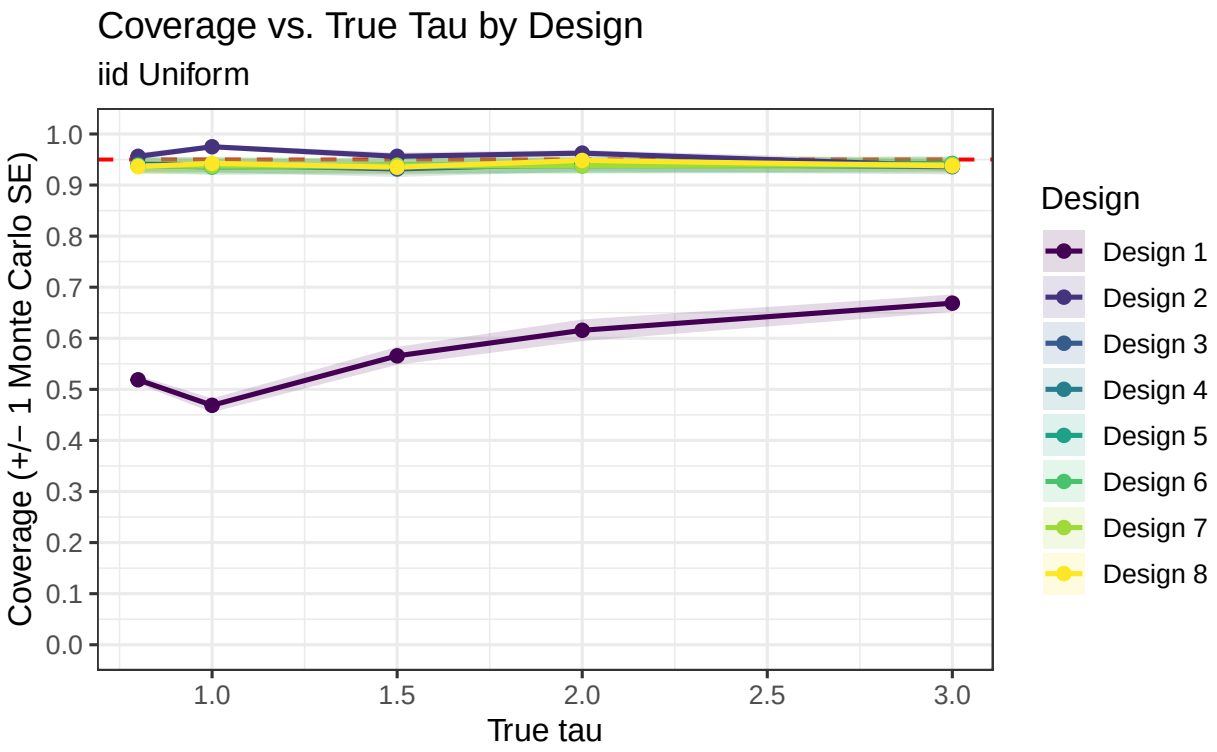

Figure 39: MLE coverage by design across tau values (iid Uniform). Nominal 95% reference shown. Valid designs maintain coverage independent of effect size.

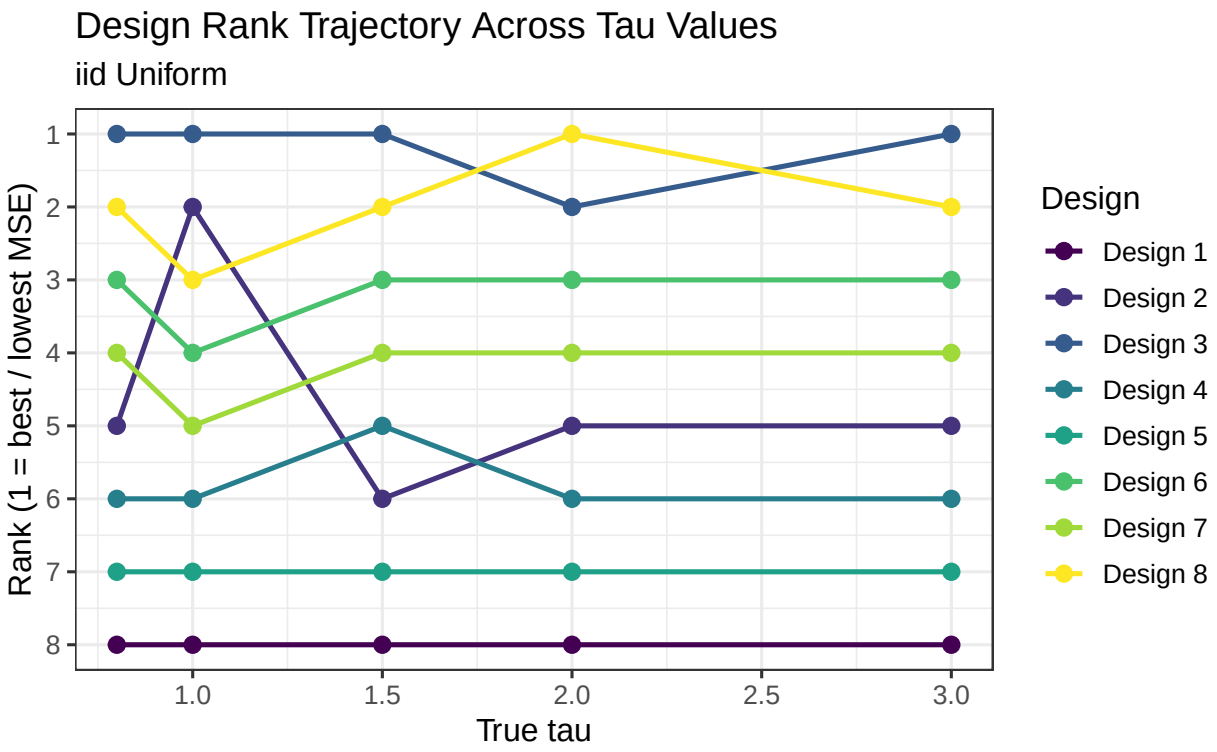

Figure 40: Design rank trajectories across tau values (rank 1 = best MSE). Stable trajectories confirm D3/D8 dominance is not an artifact of the primary effect size assumption.

```
COMPUTE Bias, SD, MSE, Coverage, Fail_Rate
STORE results row
```

---

20 Appendix B: File Locations

Table 41: RDS result files

File	Location
sim_results_DIM_combined_20260304_195321.rds	results/sim_data/
sim_results_DIM_iid_20260304_195321.rds	results/sim_data/
sim_results_DIM_poisson_20260304_195321.rds	results/sim_data/
sim_results_DIM_spatial_20260304_195321.rds	results/sim_data/
sim_results_MLE_combined_20260322_151030.rds	results/sim_data/
sim_results_MLE_iid_20260322_151030.rds	results/sim_data/
sim_results_MLE_poisson_20260322_151030.rds	results/sim_data/
sim_results_MLE_spatial_20260322_151030.rds	results/sim_data/
sim_results_MLE_tau_sweep_combined_20260408_191916.rds	results/sim_data/
sim_results_MLE_tau_sweep_iid_20260408_191916.rds	results/sim_data/
sim_results_MLE_tau_sweep_poisson_20260408_191916.rds	results/sim_data/
sim_results_MLE_tau_sweep_spatial_20260408_191916.rds	results/sim_data/

All simulation data files are located in: `results/sim_data/`

Code modules: `code/01_spatial_setup.R` through `code/08_design_recommendations.R`

Reports: `code/00_mathematical_specification.Rmd`, `code/07_results_summary.qmd`, `code/09_MLE_design_recommendation_report.qmd`